

FAB / METROLOGY / QUALIFICATION / FOUNDRY

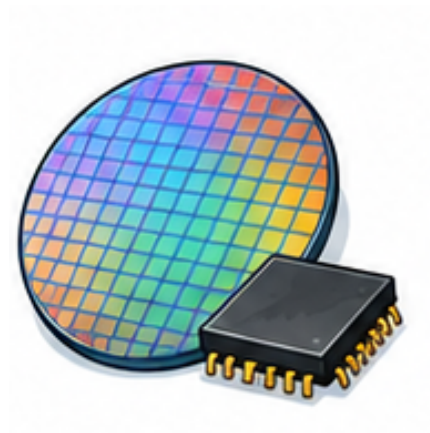
# Semiconductor English

Name the layer. Qualify the result.  
Align the handoff.

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**Eight lessons. Eleven extended conversations.**

## INSIDE THE BOOK

192 vocabulary entries / 128 reusable phrases  
220 extended-dialogue turns / 11 transfer exchanges  
Word banks and explained answer keys

Upper-intermediate to advanced (B2-C1). For process, integration, equipment, yield, product, test, and foundry-facing teams.

2026 edition

# Eight scenarios. Eight skills.

Eight semiconductor conversations connecting process context, measurement, yield, qualification, and delivery commitments.

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Back-end does not always mean packaging | Explain the current process stage and ownership without confusing wafer interconnect fabrication with package assembly.

## 02 Lithography, Reticles, and Critical Dimensions

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A three-nanometer shift without a common basis | Clarify a reported dimension change by identifying the layer, process stage, sampling, and measurement recipe before attributing a cause.

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A better center point is not a wider window | Describe a process trial improvement without extending it to untested conditions or overlooking downstream integration effects.

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Two percentage points, not a proven process gain | Report yield with counts and comparison limits, and distinguish product acceptance from evidence about process stability.

## 05 Defect Density and Cleanroom Contamination

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Twenty-four detections are not a confirmed source | Describe inspection findings with location, area, timing, and classification while avoiding premature contamination-source claims.

## 06 Equipment Uptime, Recipes, and Tool Matching

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Same recipe, different means | Distinguish recipe identity, specification performance, tool matching, and equipment availability in a transfer discussion.

## 07 Packaging, Test, and Reliability Qualification

58

One completed test is not complete qualification | Explain qualification status test by test, preserve configuration scope, and correct an overbroad customer claim.

## 08 Foundry, Tape-Out, PDK, and Capacity Communication

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A possible slot is not tape-out readiness | Negotiate a requested date by separating design sign-off, data delivery, foundry acceptance, and capacity confirmation.

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# A number needs its process context.

Semiconductor discussions compress complex evidence into phrases such as CD shift, yield gain, matched tool, and tape-out ready. Useful communication restores the missing context: layer, stage, lot, measurement recipe, test coverage, or decision status. These cases practice precise technical questions and concise updates that help different specialties work from the same facts.

## Locate the observation

Identify the layer, process stage, lot or wafer, tool, and time relevant to the claim. A change after etch is not automatically the same as a change measured in developed resist.

"The reported shift is on M2 after etch; the measurement recipe also changed."

## Keep the denominator visible

Yield and defect figures depend on what was counted, sampled, and compared. State the population and measurement basis before interpreting the difference.

"The pilot result is 92 passing die out of 100 tested, with a different product mix."

## Separate settings from performance

Identical recipe names do not prove tools behave alike. Compare the evidence against a defined matching criterion and distinguish availability from qualification.

"Both means are within specification, but the pair misses our matching limit."

## Do not compress pending into complete

A finished test, a submitted layout, and a possible capacity slot each have a limited meaning. Name the remaining checks and who can authorize the next step.

"One qualification test passed; two required tests remain open."

### PROFESSIONAL BOUNDARIES

Original fictional English practice, not a fabrication recipe, equipment procedure, product qualification, design sign-off, or production authorization. Numbers, lots, organizations, and schedules are invented. Real work must follow approved process documents, safety rules, applicable standards, customer requirements, and authorized engineering decisions.

# Back-end does not always mean packaging

## A Read the situation

**Explain the current process stage and ownership without confusing wafer interconnect fabrication with package assembly.**

New product coordinator Eric describes lot L24 as being in packaging because the status says BEOL. Integration engineer Min explains that this lot is still in wafer fabrication, building on-wafer interconnect layers. Its current operation concerns metal layer M2. In this fictional route, an inspection review must be completed before the next authorized operation. Packaging has not started, and the lot is on hold pending that review. Eric must correct the customer update without inventing a shipping date.

PEOPLE / Eric: Product coordinator | Min: Process integration engineer

### EXPAND THE ABBREVIATION AT A HANDOFF

Back-end can refer to different stages in different teams. Use the full process name when crossing organizational boundaries, then identify the actual operation and lot status. Familiar shorthand is efficient only when the receiver shares its meaning.

## Briefing checks

Circle one answer in each item. Use the briefing.

1. Where is L24 now?

- A In completed package assembly
- C In customer field qualification

- B In final shipment inspection
- D In wafer fabrication at an M2-related operation

2. What does BEOL mean in this case?

- A Back end of line, forming on-wafer interconnects
- C A confirmed customer delivery date

- B Completed assembly and final test
- D A shipping release status

3. What prevents the next operation?

- A A customer order cancellation
- C A pending inspection review and lot hold

- B A recorded shipment approval
- D A completed packaging qualification

Answers and explanations: [page 85](#).

# The vocabulary that does the work

## B Find the words / 1 of 2

Read the definitions and say the common word combinations aloud.

### wafer

A semiconductor substrate on which many devices are fabricated.

**IN USE process a wafer**

### die

An individual circuit area or chip separated from a wafer.

**IN USE identify the die**

### fab

A semiconductor fabrication facility.

**IN USE move material through the fab**

### wafer fabrication

Processes that form devices and interconnects on wafers.

**IN USE track wafer fabrication**

### process flow

The defined sequence of manufacturing operations.

**IN USE review the process flow**

### process integration

Coordination of process steps into a functioning device flow.

**IN USE assess process integration**

### FEOL

Front end of line, principally the formation of active devices.

**IN USE complete FEOL processing**

### MOL

Middle of line, connecting devices to the interconnect structure.

**IN USE review MOL interfaces**

### BEOL

Back end of line, forming the on-wafer interconnect structure.

**IN USE track BEOL progress**

### interconnect

A conductive structure connecting circuit elements.

**IN USE form an interconnect**

### metal layer

A defined level of conductive routing in a chip.

**IN USE identify the metal layer**

### via

A conductive connection between interconnect levels.

**IN USE inspect a via**

### Precision check

BEOL is still part of wafer fabrication in this conversation. It is not interchangeable with package assembly, although some organizations also use back-end for assembly and test. Expand the term and state the actual operation before communicating status.

# The vocabulary that does the work

## B Find the words / 2 of 2

Read the definitions and say the common word combinations aloud.

### dielectric

An electrically insulating material used between conductors.

**IN USE** deposit a dielectric

### process module

A related group of operations within the fabrication flow.

**IN USE** qualify a process module

### lot

A tracked group of wafers or units processed under defined identity.

**IN USE** identify the lot

### wafer ID

The identifier assigned to an individual wafer.

**IN USE** confirm the wafer ID

### route

The controlled sequence assigned to a tracked product or lot.

**IN USE** follow the approved route

### traveler

A record accompanying material through its required operations.

**IN USE** update the traveler

### WIP

Work in process: material not yet through the defined manufacturing flow.

**IN USE** report WIP status

### lot hold

A restriction preventing a lot from proceeding without disposition.

**IN USE** place a lot on hold

### disposition

An authorized decision about the next treatment or status of material.

**IN USE** obtain lot disposition

### wafer sort

Electrical testing of individual die while on the wafer.

**IN USE** schedule wafer sort

### dicing

Separating individual die from a wafer.

**IN USE** complete wafer dicing

### package assembly

Integrating and protecting die in the intended package structure.

**IN USE** begin package assembly

### Precision check

A lot at a later fabrication stage is not necessarily ready to ship. L24 remains on hold for inspection review. Its next move depends on the approved route and authorized disposition; no packaging start or delivery commitment is supplied.

# Phrases you can actually use

## **C** Notice the language / 1 of 2

### CLARIFY THE ABBREVIATION

"Here BEOL means back end of line."

### NAME THE LAYER

"The current operation concerns metal layer M2."

### EXPLAIN INTEGRATION

"We need to preserve compatibility between successive process modules."

### CHECK THE ROUTE

"What is the next operation on the approved route?"

### LOCATE THE WORK

"The lot is still in wafer fabrication."

### SEPARATE THE STAGES

"On-wafer interconnect formation is not package assembly."

### IDENTIFY THE MATERIAL

"Which lot and wafer IDs does the update cover?"

### STATE THE HOLD

"L24 is on hold pending inspection review."

---

### Back-end

May mean BEOL or assembly and test; clarify context.

### Complete

Specify the operation and whether inspection or release is included.

### Lot

A tracking group, not necessarily one die or one finished package.

# Phrases you can actually use

## **C** Notice the language / 2 of 2

### KEEP AUTHORITY EXPLICIT

"The next move needs authorized disposition."

### DISTINGUISH THE TESTS

"Wafer sort and final packaged-device test are different steps."

### CORRECT THE UPDATE

"Packaging has not started for this lot."

### ASK FOR THE STATUS BASIS

"Does completed mean processed, inspected, or released?"

### AVOID A SHIPPING INFERENCE

"This stage does not establish a shipment date."

### CONFIRM THE HANDOFF

"Which team owns the inspection review?"

### PRESERVE TRACEABILITY

"Keep the traveler linked to the lot and operation."

### CLOSE PRECISELY

"Report the stage, hold reason, owner, and next decision."

---

### Next

Must follow the approved route and current restrictions.

### Ready

Needs a named next step and its prerequisites.

### Released

Means a specific authorized status change, not merely elapsed time.

# Choose the precise message

## **D** Check your understanding

Circle one best answer for each item. Use the stated purpose and case facts, not the option that merely sounds most polite.

### 1. Which customer update is accurate?

- A** L24 remains in wafer fabrication at M2, on hold for inspection review; packaging has not started.
- B** L24 has completed the interconnect stack and awaits wafer sort.
- C** L24 is in package assembly with final electrical test pending.
- D** L24 is at M2 and can move while the inspection report is being completed.

### 2. Why expand BEOL for the coordinator?

- A** BEOL always means package assembly in every organization.
- B** Acronyms eliminate the need to name an operation.
- C** The lot ID alone reveals its full status.
- D** Different teams can use back-end for different parts of the manufacturing chain.

### 3. What is the appropriate next-step wording?

- A** Promise shipment before checking the remaining flow.
- B** The lot awaits inspection review and authorized disposition under its route.
- C** Move the lot because M2 is usually near completion.
- D** Skip the review because packaging has not started.

### 4. Which distinction is correct?

- A** Wafer sort tests die on the wafer; package assembly integrates die into packages.
- B** Wafer sort separates die mechanically; dicing classifies their electrical performance.
- C** BEOL joins die to package substrates; package assembly forms on-wafer metal layers.
- D** FEOL forms package connections; BEOL forms the active transistor devices.

[Answers and explanations: page 85.](#)

# Back-end does not always mean packaging

## E Conversations / turns 1-10

Complete the numbered gaps. Use each bank entry once, without changing its form. Cover the answers at the end until you finish.

### WORD + PHRASE BANK

process flow / disposition / FEOL / traveler / wafer fabrication / Wafer sort / metal layer / package assembly / lot hold / BEOL

- 01 **Eric:** I wrote that L24 is in packaging because the status says BEOL. The customer wants to know whether that means finished parts will be ready soon.
- 02 **Min:** Here <sup>1</sup> \_\_\_\_\_ means back end of line. We are building the interconnect structure on the wafer, so your update places the lot at a later stage than the record supports.
- 03 **Eric:** That explains the confusion. The assembly team also calls its work back-end, so I assumed both groups meant the same stage.
- 04 **Min:** Use <sup>2</sup> \_\_\_\_\_ in the customer message, then name the actual operation. Expanding the term is especially useful when the reader does not share our process shorthand.
- 05 **Eric:** I will name M2 in the update. Does that identify the current layer, rather than mean the full interconnect stack is finished?
- 06 **Min:** Correct. M2 is the relevant <sup>3</sup> \_\_\_\_\_. A layer name locates the work, but it does not by itself tell you that inspection, disposition, or later stages have finished.
- 07 **Eric:** Can you explain how that fits with the earlier device work? I need enough context to brief the customer without reciting every operation in the fab.
- 08 **Min:** <sup>4</sup> \_\_\_\_\_ principally forms the active devices, while later interconnect work connects them. The exact flow is technology-specific, so use our product route for the actual sequence and status.
- 09 **Eric:** I also see that inspection review is pending. The lot cannot simply move to its next listed operation because someone has already processed the current step.
- 10 **Min:** It is on a <sup>5</sup> \_\_\_\_\_. That restriction remains in place until the required review and authorized decision. Processed, inspected, and released are different status claims.

# The conversation, continued

## E Conversations / turns 11-20

Continue with the word bank on page 10.

- 11 **Eric:** Then the update should identify what must happen before the next move, not describe the hold as if it were just a routine scheduling delay.
- 12 **Min:** Yes. We need <sup>6</sup> \_\_\_\_\_ under the approved process. The review owner can explain the decision basis; we should not predict the outcome before that evidence is assessed.
- 13 **Eric:** I will confirm the owner and preserve the lot reference. We should also keep the record connected to the individual wafers where the inspection information requires that detail.
- 14 **Min:** Keep the <sup>7</sup> \_\_\_\_\_ and relevant wafer IDs aligned with the update. Otherwise two teams may discuss different material while assuming they are referring to the same issue.
- 15 **Eric:** What later stages should I distinguish in the message? I need to explain why an M2 status does not yet give the customer a finished-product date.
- 16 **Min:** Exactly. <sup>8</sup> \_\_\_\_\_, dicing, assembly, and subsequent checks have their own place in the approved route. Their timing is not established by the M2 entry alone.
- 17 **Eric:** I will correct the packaging statement now, before the customer plans around a stage the lot has not reached.
- 18 **Min:** Say that <sup>9</sup> \_\_\_\_\_ has not started for this lot. You can give a precise current status without inventing dates for the remaining operations.
- 19 **Eric:** The revised message will locate L24 in wafer fabrication at M2, explain the inspection hold, and identify the next review. It will leave shipment timing uncommitted.
- 20 **Min:** That reflects the <sup>10</sup> \_\_\_\_\_ accurately. The customer gets a clear explanation of progress and the pending decision instead of an optimistic interpretation of an abbreviation.

### ANSWERS / EXPLANATIONS: PAGE 86

1. BEOL | 2. wafer fabrication | 3. metal layer | 4. FEOL | 5. lot hold | 6. disposition | 7. traveler | 8. Wafer sort | 9. package assembly | 10. process flow

# Name the stage before the date

## F Say it

Lot N6 is forming on-wafer interconnects. It is on hold for inspection review. Package assembly has not started, and no shipping date is confirmed.

Complete the four gaps. Use each bank entry once.

### WORD + PHRASE BANK

BEOL / package assembly / lot hold / disposition

Coordinator: "I will correct the update: this on-wafer interconnect work is still

<sup>1</sup> \_\_\_\_\_."

Engineer: "Keep the restriction visible too; the record still shows a

<sup>2</sup> \_\_\_\_\_."

Coordinator: "Then I should not tell the customer we have started

<sup>3</sup> \_\_\_\_\_."

Engineer: "Correct. Ask the review owner for the authorized <sup>4</sup> \_\_\_\_\_ before discussing the next move."

### ANSWERS / EXPLANATIONS: PAGE 86

1. BEOL | 2. lot hold | 3. package assembly | 4. disposition

## Rehearse with a fixed script

1. Check the dialogue answers. Read Eric and Min aloud, expanding BEOL and FEOL when they appear.
2. Switch roles. Read turns 9-20 again, stressing the hold, required decision, and stage not yet started.
3. Complete the transfer and check its key. Read the corrected update without adding a shipping date.

### WORKING ALONE

Read both roles, pausing for two seconds when the speaker changes. Repeat with the bank covered, then check any missed expressions.

# A three-nanometer shift without a common basis

## A Read the situation

**Clarify a reported dimension change by identifying the layer, process stage, sampling, and measurement recipe before attributing a cause.**

Lithography engineer Lina reviews a reported critical-dimension increase with metrology engineer Theo. The summary initially omits the layer and stage. The records show M2 after etch, with means of 86 nm and 89 nm. However, the two measurements used different CD-SEM recipes and different sampling sites. No comparable remeasurement, specification assessment, or cause investigation is complete. Lina must report the apparent 3 nm difference without treating it as a confirmed lithography shift or changing an exposure setting.

PEOPLE / Lina: Lithography engineer | Theo: Metrology engineer

### ASK WHICH DIMENSION WAS MEASURED

CD shift sounds specific but can conceal different features, stages, or measurement settings. A useful challenge asks for the measurement basis before debating process causes. Do not assume that a post-etch result isolates lithography or that a changed mean proves a specification violation.

## Briefing checks

Circle one answer in each item. Use the briefing.

1. Which layer and stage do the records identify?

- A Unspecified resist before exposure
- C M2 before every fabrication step

- B Final package thickness
- D M2 after etch

2. What is the reported numerical difference?

- A An increase of 89 nm
- C An increase of 3 nm

- B No numerical difference
- D A decrease of 3 nm

3. Why is a direct process attribution premature?

- A The layer is now identified.
- C The measurement recipes and sampling sites differ.

- B Every post-etch result can only originate in lithography.
- D Both figures use nanometers.

Answers and explanations: page 87.

# The vocabulary that does the work

## B Find the words / 1 of 2

Read the definitions and say the common word combinations aloud.

### **lithography**

Pattern formation using exposure of a sensitive layer.

**IN USE review lithography performance**

### **reticle**

A patterned optical element used in lithographic exposure.

**IN USE identify the reticle**

### **photoresist**

A radiation-sensitive material used to form a pattern.

**IN USE coat the photoresist**

### **exposure dose**

The delivered exposure energy per unit area.

**IN USE record exposure dose**

### **focus offset**

A displacement from a defined best-focus reference.

**IN USE evaluate focus offset**

### **critical dimension**

A feature dimension important to process or device performance, abbreviated CD.

**IN USE measure critical dimension**

### **CD-SEM**

A scanning electron microscope used for critical-dimension measurement.

**IN USE verify the CD-SEM recipe**

### **measurement recipe**

The controlled settings and analysis used to obtain a measurement.

**IN USE confirm the measurement recipe**

### **sampling site**

A specified location selected for measurement.

**IN USE align sampling sites**

### **post-develop**

Measured or occurring after resist development.

**IN USE review post-develop CD**

### **post-etch**

Measured or occurring after an etch operation.

**IN USE compare post-etch measurements**

### **etch bias**

A dimensional difference associated with pattern transfer through etch.

**IN USE evaluate etch bias**

### **Precision check**

The arithmetic difference is 3 nm. That does not prove a 3 nm physical process shift because the recipes and sampled sites differ. A post-etch CD combines the history of pattern formation and transfer; it does not identify exposure as the sole cause.

# The vocabulary that does the work

## B Find the words / 2 of 2

Read the definitions and say the common word combinations aloud.

### overlay

Alignment accuracy between patterned layers.

**IN USE** measure overlay

### alignment mark

A feature used to position a layer or measurement.

**IN USE** inspect alignment marks

### line-edge roughness

Variation of a patterned line edge from its intended shape.

**IN USE** assess line-edge roughness

### line-width roughness

Variation in the width of a patterned line.

**IN USE** assess line-width roughness

### pitch

The repeat distance between corresponding features.

**IN USE** specify feature pitch

### pattern fidelity

Agreement between the intended and produced pattern.

**IN USE** assess pattern fidelity

### optical proximity correction

Layout adjustments compensating for imaging effects, abbreviated OPC.

**IN USE** review optical proximity correction

### focus-exposure matrix

A test pattern set spanning selected focus and dose conditions.

**IN USE** analyze a focus-exposure matrix

### depth of focus

A usable focus range under a defined imaging criterion.

**IN USE** assess depth of focus

### exposure latitude

A usable dose range under a defined imaging criterion.

**IN USE** estimate exposure latitude

### measurement bias

A systematic measurement difference relative to a reference.

**IN USE** assess measurement bias

### remeasurement

A repeated measurement under identified conditions.

**IN USE** request comparable remeasurement

### Precision check

CD and overlay measure different properties: feature size and layer alignment. Neither should be substituted for the other. A specification judgment also needs the applicable limits and measurement basis, which are not supplied as resolved facts in this case.

# Phrases you can actually use

## **C** Notice the language / 1 of 2

### LOCATE THE DIMENSION

"Which layer and feature does the CD refer to?"

### STATE THE NUMBERS

"The reported means are 86 and 89 nm."

### ASK ABOUT MEASUREMENT

"Did both datasets use the same CD-SEM recipe?"

### AVOID A CAUSAL SHORTCUT

"Post-etch CD does not isolate exposure performance."

### IDENTIFY THE STAGE

"These measurements are from M2 after etch."

### QUALIFY THE DIFFERENCE

"The apparent difference is 3 nm."

### CHECK SAMPLING

"Were the same types of sites sampled?"

### SEPARATE METRICS

"Overlay is not another name for critical dimension."

---

## Shift

Specify whether apparent, confirmed, or attributed.

## CD

Identify feature, layer, stage, and measurement method.

## Post-etch

Describes a stage, not proof of which earlier step caused a change.

# Phrases you can actually use

## C Notice the language / 2 of 2

### REQUEST COMPARABILITY

"Remeasure on an agreed, traceable basis."

### AVOID AN UNSUPPORTED ADJUSTMENT

"Do not infer an exposure change from this comparison alone."

### DISTINGUISH VARIATION

"A mean shift and within-wafer variation are different questions."

### DESCRIBE THE NEXT CHECK

"Resolve the measurement difference before attributing the process cause."

### PRESERVE THE ORIGINALS

"Keep both original recipes with their datasets."

### ASK FOR LIMITS

"Which specification applies to this feature and stage?"

### CLARIFY RETICLE IDENTITY

"Which reticle revision was used for the layer?"

### CLOSE THE REPORT

"State the apparent shift and the comparability limitation together."

---

### Same recipe

Confirm the actual revision, not only the displayed name.

### Out of spec

Needs the applicable limit and a valid measurement basis.

### Adjust

Requires an authorized technical decision, not a language exercise inference.

# Choose the precise message

## **D** Check your understanding

Circle one best answer for each item. Use the stated purpose and case facts, not the option that merely sounds most polite.

### 1. Which report sentence is justified?

- A** Both datasets prove the same physical result because they use nanometers.
- B** M2 is out of specification without any stated limit.
- C** The means differ by 3 nm, but recipe and site changes limit comparability.
- D** Exposure drift has been confirmed at exactly 3 nm.

### 2. Which comparison best helps investigate the apparent change?

- A** Increasing sample counts while leaving recipe and site differences uncontrolled
- B** Comparing the post-etch mean with a post-develop mean as though the stages were identical
- C** Comparable measurements on identified sites using an agreed recipe basis
- D** Repeating both datasets with their different original recipes and locations

### 3. Why is post-etch important?

- A** It makes the sampling locations irrelevant.
- B** The result follows pattern transfer and does not isolate lithography alone.
- C** It means no etch process has occurred.
- D** It establishes that every cause is in the reticle.

### 4. Which distinction is accurate?

- A** CD concerns feature size; overlay concerns layer alignment.
- B** CD measures layer alignment; overlay measures a line's width.
- C** Pitch measures line-edge variation; line-edge roughness measures repeat distance.
- D** Etch bias is the exposure energy; exposure dose is the post-etch dimensional change.

Answers and explanations: page 87.

# A three-nanometer shift without a common basis

## E Conversations / turns 1-10

Complete the numbered gaps. Use each bank entry once, without changing its form. Cover the answers at the end until you finish.

### WORD + PHRASE BANK

pattern fidelity / Overlay / remeasurement / sampling site / reticle / critical dimension / measurement recipe / post-etch / Etch bias / measurement bias

- 01 **Lina:** The review slide says CD shifted, but it does not identify the layer or the measurement stage. I cannot tell which process team should investigate first.
- 02 **Theo:** The records identify <sup>1</sup> \_\_\_\_\_ M2 measurements. The two means are eighty-six and eighty-nine nanometers, so the reported numerical increase is three nanometers.
- 03 **Lina:** That locates the result, but I still need to know whether the two measurements were obtained on a comparable basis before calling it a physical process shift.
- 04 **Theo:** The <sup>2</sup> \_\_\_\_\_ changed between datasets. Both used CD-SEM, but sharing the instrument type does not mean their settings and analysis were identical.
- 05 **Lina:** Were the same locations measured? A different selection across the wafer could also change the average without establishing the process explanation stated in the meeting.
- 06 **Theo:** The <sup>3</sup> \_\_\_\_\_ selection changed too. We need the location records alongside the recipes, rather than treating the two mean values as a controlled before-and-after comparison.
- 07 **Lina:** I will keep the three-nanometer difference and remove the exposure-drift conclusion. Can we show the changed recipe and sites beside the two means?
- 08 **Theo:** Correct. The <sup>4</sup> \_\_\_\_\_ result is important, but the present comparison does not separate process behavior from measurement and sampling effects.
- 09 **Lina:** Because this is after etch, the final feature can reflect both the earlier resist pattern and the transfer step. It is not a direct exposure-only measurement.
- 10 **Theo:** Exactly. <sup>5</sup> \_\_\_\_\_ is a separate consideration in the investigation. Naming it does not establish that etch caused this case, but it prevents an unjustified shortcut to lithography.

# The conversation, continued

## E Conversations / turns 11-20

Continue with the word bank on page 19.

- 11 **Lina:** The review also asks whether this is an alignment problem. Can we clarify that separately? I do not want the CD figure copied into an overlay conclusion.
- 12 **Theo:** <sup>6</sup> \_\_\_\_\_ concerns alignment between layers. It is not another name for feature width, so this CD comparison cannot be relabeled as an overlay finding.
- 13 **Lina:** Our next request should identify the relevant feature, layer, stage, locations, and agreed measurement settings. Then the engineers can compare results with a traceable basis.
- 14 **Theo:** Request comparable <sup>7</sup> \_\_\_\_\_ through the approved process. Preserve the original datasets too, because their recipes and locations are part of the explanation for what was initially reported.
- 15 **Lina:** If the difference persists, the team can investigate possible process causes with better evidence. If it does not, we still need to explain the original discrepancy.
- 16 **Theo:** Yes. Assess <sup>8</sup> \_\_\_\_\_ and other relevant effects rather than force either outcome. The aim is a supported interpretation, not a preferred conclusion that the process changed or stayed identical.
- 17 **Lina:** I will include the reticle revision in the context record as well. That gives the investigation the configuration information without implying the reticle has been found defective.
- 18 **Theo:** Good. The <sup>9</sup> \_\_\_\_\_ identity belongs with the layer history. Configuration details help evaluate hypotheses, but listing a component does not make it the cause.
- 19 **Lina:** The revised slide will keep the two means and the three-nanometer difference, state the changed measurement basis, and leave the cause and specification assessment open.
- 20 **Theo:** That protects <sup>10</sup> \_\_\_\_\_ discussions from unsupported conclusions. We can plan the next evidence review without treating this comparison as permission to change exposure settings.

### ANSWERS / EXPLANATIONS: PAGE 88

1. post-etch | 2. measurement recipe | 3. sampling site | 4. critical dimension | 5. Etch bias | 6. Overlay | 7. remeasurement | 8. measurement bias | 9. reticle | 10. pattern fidelity

# Same units, different measurement basis

## F Say it

A layer has reported means of 120 and 124 nm after etch. Different measurement recipes and sampling sites were used. The cause is not established.

Complete the four gaps. Use each bank entry once.

### WORD + PHRASE BANK

sampling sites / overlay / measurement recipe / critical dimension

Engineer: "Check that both reports refer to the same <sup>1</sup> \_\_\_\_\_."

Reviewer: "Send the settings and analysis from each <sup>2</sup> \_\_\_\_\_."

Engineer: "I will attach a wafer map of both sets of <sup>3</sup> \_\_\_\_\_."

Reviewer: "Good. An alignment claim would need separate <sup>4</sup> \_\_\_\_\_ evidence."

### ANSWERS / EXPLANATIONS: PAGE 88

1. critical dimension | 2. measurement recipe | 3. sampling sites | 4. overlay

## Rehearse with a fixed script

1. Check the answers. Read 86 nm and 89 nm as complete spoken measurements.
2. Switch roles for turns 7-16. Stress apparent difference, post-etch, and the changed measurement basis.
3. Check and read the transfer twice. Keep feature size distinct from alignment.

### WORKING ALONE

Read both roles, pausing for two seconds when the speaker changes. Repeat with the bank covered, then check any missed expressions.

# A better center point is not a wider window

## A Read the situation

**Describe a process trial improvement without extending it to untested conditions or overlooking downstream integration effects.**

Deposition engineer Yuki and integration engineer Sam review a trial film process. At the nominal setting, a defined thickness-nonuniformity metric falls from 5% to 3%; lower is better on this unchanged measurement basis. Only that center condition has been tested. The proposed low and high settings remain untested, and effects on subsequent etch and chemical mechanical planarization are not assessed. The new recipe is not released. A summary nevertheless says the complete process window is validated.

PEOPLE / Yuki: Deposition engineer | Sam: Integration engineer

### CELEBRATE A BOUNDED IMPROVEMENT

An encouraging center-point result is worth reporting. State the metric and tested condition, then identify the untested range and integration questions. This lets colleagues recognize progress without confusing a promising experiment with a released process window.

## Briefing checks

Circle one answer in each item. Use the briefing.

1. What improved in the supplied trial?

**A** The new recipe received production release.

**B** The stated nonuniformity metric fell from 5% to 3% at the nominal setting.

**C** Every condition in the proposed range was tested.

**D** Etch and planarization were fully qualified.

2. What is the absolute change in that percentage metric?

**A** A decrease of 2 percentage points

**B** A decrease of 2 relative percent

**C** An increase of 3 percentage points

**D** A decrease of 5 percentage points

3. Which conclusion is premature?

**A** The low and high settings remain untested.

**B** Downstream integration effects remain open.

**C** The complete process window is validated.

**D** The nominal-setting result improved on the stated metric.

Answers and explanations: [page 89](#).

# The vocabulary that does the work

## B Find the words / 1 of 2

Read the definitions and say the common word combinations aloud.

### deposition

Formation of a material film on a surface.

**IN USE** qualify a deposition step

### thin film

A material layer with a small controlled thickness.

**IN USE** characterize the thin film

### CVD

Chemical vapor deposition, forming a film through chemical reactions of vapor precursors.

**IN USE** evaluate CVD film properties

### PVD

Physical vapor deposition, forming a film from material vaporized by methods such as sputtering or evaporation.

**IN USE** review PVD performance

### ALD

Atomic layer deposition, using successive self-limiting surface reactions.

**IN USE** assess ALD conformity

### precursor

A starting substance used in a film-forming reaction.

**IN USE** qualify the precursor

### conformality

How consistently a film coats different surface orientations or features.

**IN USE** assess film conformity

### step coverage

Film thickness on a specified feature relative to thickness on a reference surface.

**IN USE** evaluate step coverage

### film stress

Internal stress within a deposited film.

**IN USE** measure film stress

### thickness nonuniformity

Variation in film thickness under a defined metric.

**IN USE** report thickness nonuniformity

### etch selectivity

Relative removal rates of different materials in an etch process.

**IN USE** assess etch selectivity

### anisotropy

Direction-dependent behavior, such as preferential vertical removal.

**IN USE** evaluate etch anisotropy

### Precision check

The metric improves by 2 percentage points, a 40% relative reduction from its initial 5% value. Neither calculation establishes the proposed window. The same measurement definition is used here; other nonuniformity formulas must not be mixed into the comparison.

# The vocabulary that does the work

## B Find the words / 2 of 2

Read the definitions and say the common word combinations aloud.

### endpoint detection

Monitoring used to identify a process completion condition.

**IN USE verify endpoint detection**

### overetch

Additional etching beyond a defined clearing or endpoint condition.

**IN USE assess overetch effects**

### CMP

Chemical mechanical planarization, combining chemical and mechanical surface removal.

**IN USE evaluate CMP compatibility**

### slurry

A polishing fluid mixture supplying chemical action and often suspended abrasive particles in CMP.

**IN USE qualify the CMP slurry**

### dishing

Local depression of a material surface during planarization.

**IN USE measure dishing**

### erosion

Excess regional material removal during planarization.

**IN USE assess erosion**

### process window

A range of conditions meeting defined process requirements.

**IN USE establish the process window**

### nominal setting

The chosen central or reference process setting.

**IN USE test the nominal setting**

### corner condition

A selected limiting combination in an evaluation range.

**IN USE evaluate corner conditions**

### process interaction

An effect arising from the relationship between process steps or factors.

**IN USE investigate process interactions**

### integration trade-off

An exchange between benefits and effects across process steps.

**IN USE explain the integration trade-off**

### recipe release

Authorization of a controlled process recipe for defined use.

**IN USE obtain recipe release**

### Precision check

A deposition improvement can affect later etch and planarization behavior. The case supplies no result for those steps. Describe their assessment as pending, not as either a guaranteed benefit or a demonstrated new failure.

# Phrases you can actually use

## C Notice the language / 1 of 2

03

### IDENTIFY THE METRIC

"The defined thickness-nonuniformity metric fell from 5% to 3%."

### STATE THE CONDITION

"The result applies to the nominal setting only."

### LIMIT THE CLAIM

"One center point does not establish a process window."

### CHECK INTERACTIONS

"What does the film change mean for the subsequent etch?"

### KEEP THE BASIS FIXED

"Both values use the same measurement definition."

### SEPARATE CHANGES

"That is two percentage points, or a 40% relative reduction."

### NAME THE MISSING TESTS

"The proposed low and high settings are untested."

### ASK ABOUT PLANARIZATION

"Has compatibility with the CMP step been assessed?"

---

## Uniformity

State the formula or metric, not only a percentage.

## Window

A range meeting requirements, not one successful setting.

## Corner

A defined edge or combination in the evaluation plan.

# Phrases you can actually use

## C Notice the language / 2 of 2

03

### AVOID A CAUSAL LEAP

"Better uniformity does not prove every film property improved."

### QUALIFY MATURITY

"This is a trial result, not a released recipe."

### PRESERVE CONFIGURATION

"Keep the trial recipe revision with the measurement record."

### AVOID DECLARING FAILURE

"Untested corners are unresolved, not failed."

### IDENTIFY TRADE-OFFS

"We still need the integration trade-off assessment."

### ASK FOR CRITERIA

"Which responses must meet limits across the range?"

### REQUEST BOUNDED FOLLOW-UP

"Assess the planned range and the identified downstream effects."

### CLOSE THE SUMMARY

"Report the center-point result and the remaining validation scope."

---

### Better

Name the response that improved and what was not assessed.

### Integrated

Requires consideration of connected process steps.

### Released

An authorization status, not a synonym for promising.

# Choose the precise message

## **D** Check your understanding

Circle one best answer for each item. Use the stated purpose and case facts, not the option that merely sounds most polite.

### 1. Which summary preserves the evidence boundary?

- A** The trial establishes the window, with etch and CMP review needed only after release.
- B** The nominal-setting metric improved; the range and downstream effects remain untested.
- C** The metric improved at nominal, so the same reduction applies at the low and high settings.
- D** A lower nonuniformity metric establishes improved film stress and conformality too.

### 2. What is the relative reduction from 5% to 3%?

- A** 60%
- B** 3%
- C** 40%
- D** 2%

### 3. Why review downstream etch and CMP?

- A** Lower thickness variation removes the need to assess other film properties.
- B** A film change may alter later process behavior.
- C** The deposition metric already establishes the etch selectivity.
- D** An unchanged CMP recipe proves the new film polishes identically.

### 4. Which status should remain separate from trial success?

- A** The recorded original value of 5%
- B** Recipe release
- C** The reported nominal-setting value of 3%
- D** The unchanged metric definition

Answers and explanations: [page 89](#).

# A better center point is not a wider window

## E Conversations / turns 1-10

Complete the numbered gaps. Use each bank entry once, without changing its form. Cover the answers at the end until you finish.

### WORD + PHRASE BANK

integration trade-off / CMP / recipe release / thickness nonuniformity / process window / process interaction / deposition / etch selectivity / corner condition / nominal setting

- 01 **Yuki:** The nominal-setting trial reduced our thickness-nonuniformity metric from five percent to three percent. The summary now says the full process window is validated, which seems too broad.
- 02 **Sam:** It is too broad. A <sup>1</sup> \_\_\_\_\_ covers a range of conditions meeting defined requirements. We have one encouraging center-point result, not evidence for the entire proposed range.
- 03 **Yuki:** Both values use the same measurement definition, and lower is better for this metric. I want to report that improvement clearly without losing the limitation.
- 04 **Sam:** Name <sup>2</sup> \_\_\_\_\_ explicitly. Saying the process improved could imply that every relevant response was assessed, while the supplied result addresses only this defined metric.
- 05 **Yuki:** The absolute decrease is two percentage points. Relative to the original five percent, the decrease is forty percent, not merely two percent.
- 06 **Sam:** Keep that arithmetic tied to the <sup>3</sup> \_\_\_\_\_. Neither way of expressing the improvement extends the evidence to settings that have not been run.
- 07 **Yuki:** I will mark the low and high settings untested. What evidence does the approved plan require before we can make a claim about that range?
- 08 **Sam:** Correct. Each planned <sup>4</sup> \_\_\_\_\_ needs its own assessment under the approved plan. An untested condition is unresolved, not a favorable result or a demonstrated defect.
- 09 **Yuki:** The deposition change may also affect other film characteristics. A lower thickness-variation metric does not tell us that stress, coverage, or later pattern transfer remained acceptable.
- 10 **Sam:** That is an <sup>5</sup> \_\_\_\_\_ question. We need to assess the relevant responses and interactions rather than optimize one number in isolation from the product flow.

# The conversation, continued

## E Conversations / turns 11-20

03

Continue with the word bank on page 28.

- 11 **Yuki:** For the subsequent etch, the team has not yet checked how the changed film behaves relative to the neighboring materials. That could affect the interpretation of the trial.
- 12 **Sam:** Review <sup>6</sup> \_\_\_\_\_ where it matters to the approved integration assessment. Naming the concern identifies a question to investigate, not a result we can claim already occurred.
- 13 **Yuki:** The planarization team also needs the proposed film information. Their review has not started, so I should not imply that the next surface-finishing step is already covered.
- 14 **Sam:** State that <sup>7</sup> \_\_\_\_\_ compatibility remains open. The deposition result does not establish that planarization behavior, surface effects, or related requirements are unchanged.
- 15 **Yuki:** We can identify the relevant responses in the evaluation plan without improvising operating settings during this review. The process owners should define the approved technical work.
- 16 **Sam:** Yes. Capture each <sup>8</sup> \_\_\_\_\_ question and its owner. That makes the remaining work specific enough to schedule and review without treating the meeting as recipe authorization.
- 17 **Yuki:** Please keep trial revision on the summary. Planning is asking about routine use, and I do not want this result mistaken for a production release.
- 18 **Sam:** Keep <sup>9</sup> \_\_\_\_\_ separate from the experiment. A favorable metric can support the review, but the required scope and approvals still need to be completed.
- 19 **Yuki:** I will revise the summary to report the center-point improvement, the unchanged metric basis, and the untested range. I will list the etch and planarization reviews as pending.
- 20 **Sam:** That presents the <sup>10</sup> \_\_\_\_\_ result accurately. The team can recognize progress while seeing exactly what evidence is still needed before making a broader process claim.

### ANSWERS / EXPLANATIONS: PAGE 90

1. process window | 2. thickness nonuniformity | 3. nominal setting | 4. corner condition | 5. integration trade-off | 6. etch selectivity | 7. CMP | 8. process interaction | 9. recipe release | 10. deposition

# An improved metric with a limited range

## F Say it

At one nominal setting, an unchanged nonuniformity metric falls from 8% to 6%. Lower is better. Edge settings and downstream etch remain untested; the recipe is not released.

Complete the four gaps. Use each bank entry once.

### WORD + PHRASE BANK

process window / integration question / recipe release / nominal setting

Engineer: "The metric fell from eight to six percent, but only at the

1 \_\_\_\_\_."

Reviewer: "Keep the low and high settings open; we have not established the

2 \_\_\_\_\_."

Engineer: "I will also list the untested effect on the later etch as an

3 \_\_\_\_\_."

Reviewer: "And keep routine production use pending the authorized

4 \_\_\_\_\_."

### ANSWERS / EXPLANATIONS: PAGE 90

1. nominal setting | 2. process window | 3. integration question | 4. recipe release

## Rehearse with a fixed script

1. Check the answers. Read the opening six turns, distinguishing two percentage points from forty percent.

2. Switch roles. Read the etch and CMP exchanges while keeping both assessments pending.

3. Complete and check the transfer. Read nominal setting, process window, and release as separate claims.

### WORKING ALONE

Read both roles, pausing for two seconds when the speaker changes. Repeat with the bank covered, then check any missed expressions.

# Two percentage points, not a proven process gain

04

## A Read the situation

**Report yield with counts and comparison limits, and distinguish product acceptance from evidence about process stability.**

Yield engineer Ravi presents a pilot lot with 92 passing die out of 100 tested, compared with a reference lot with 900 passing die out of 1,000 tested. The product mixes differ, and no controlled comparison or statistical significance assessment has been completed. Manager Elena wants to announce that a process change caused a two-percent yield improvement. The team also has no time-ordered control-chart analysis supporting a stability claim. Ravi must preserve the useful observations while correcting the causal, percentage, and stability language.

PEOPLE / Elena: Yield program manager | Ravi: Yield engineer

### GIVE THE COUNTS BEFORE THE CONCLUSION

A favorable yield difference can start an investigation without ending it. State the passing and tested counts, identify changed product mix, and distinguish an absolute percentage-point difference from a relative percentage change. Avoid using the words significant or stable as casual substitutes for better.

## Briefing checks

Circle one answer in each item. Use the briefing.

1. What is the pilot yield?

- A 100% because the pilot completed
- B 92% from 92 of 100 tested die
- C 90% from 900 of 1,000 die
- D 2% from two lots

2. What is the absolute difference between the observed yields?

- A Exactly 2 relative percent
- B 20 percentage points
- C 2 percentage points
- D 2 passing die

3. Which factor limits the causal comparison?

- A The pilot is called a pilot.
- B The reference contains more passing die in absolute number.
- C The product mixes differ.
- D Both yields have denominators.

Answers and explanations: page 91.

# The vocabulary that does the work

## B Find the words / 1 of 2

Read the definitions and say the common word combinations aloud.

### metrology

The science and practice of measurement.

**IN USE review metrology consistency**

### electrical yield

The proportion meeting defined electrical test criteria.

**IN USE report electrical yield**

### pass count

The number of tested items meeting the stated criteria.

**IN USE preserve the pass count**

### tested population

The set of items included in a reported test result.

**IN USE define the tested population**

### denominator

The total used as the base of a fraction or percentage.

**IN USE state the denominator**

### product mix

The composition of different products in a dataset.

**IN USE adjust for product mix**

### pilot lot

A limited production lot used to evaluate a process or product.

**IN USE evaluate a pilot lot**

### reference lot

A lot used as a stated comparison basis.

**IN USE identify the reference lot**

### percentage point

A unit for the absolute difference between percentages.

**IN USE report percentage-point change**

### relative change

A change divided by its specified starting value.

**IN USE calculate relative change**

### confounding factor

A factor that prevents clean attribution of an observed relationship.

**IN USE identify a confounding factor**

### stratification

Separating data into relevant comparable groups.

**IN USE stratify by product**

### Precision check

The pilot has 92% yield and the reference 90%. The difference is 2 percentage points, or about 2.22% relative to 90%. Different product mixes and sample sizes mean this observation alone does not establish a causal improvement or statistical significance.

# The vocabulary that does the work

## B Find the words / 2 of 2

Read the definitions and say the common word combinations aloud.

### SPC

Statistical process control, using data to monitor process behavior.

**IN USE** apply SPC

### control chart

A time-ordered chart with a defined process-monitoring method.

**IN USE** review the control chart

### control limit

A monitoring boundary derived under a stated process model or method.

**IN USE** distinguish control limits

### specification limit

A defined product or process acceptance boundary.

**IN USE** check specification limits

### common-cause variation

Variation inherent in the currently defined process system.

**IN USE** characterize common-cause variation

### special-cause variation

Variation associated with a nonroutine identifiable influence.

**IN USE** investigate special-cause variation

### control signal

A result meeting a defined rule for investigation on a control chart.

**IN USE** respond to a control signal

### process stability

Consistent statistical behavior under the defined monitoring basis.

**IN USE** assess process stability

### subgroup

A defined set of observations grouped for process analysis.

**IN USE** define rational subgroups

### measurement system

The instruments, methods, people, and conditions producing measurements.

**IN USE** assess the measurement system

### yield learning

Investigation that improves understanding of yield losses and drivers.

**IN USE** support yield learning

### causal attribution

Assigning an observed effect to a particular cause.

**IN USE** qualify causal attribution

### Precision check

Control limits describe expected process behavior under a defined monitoring method; specification limits express acceptance requirements. Passing a specification does not establish stability, and a control signal does not automatically identify either the cause or the disposition of every unit.

# Phrases you can actually use

## C Notice the language / 1 of 2

### GIVE THE PILOT COUNTS

"The pilot passed 92 of 100 tested die."

### GIVE THE REFERENCE COUNTS

"The reference passed 900 of 1,000 tested die."

### NAME THE ABSOLUTE CHANGE

"The observed difference is two percentage points."

### NAME THE RELATIVE BASIS

"Relative to 90%, that is approximately 2.22%."

### EXPOSE THE MIX

"The two lots have different product mixes."

### AVOID CAUSALITY BY TIMING

"The process change is not isolated by this comparison."

### QUALIFY SIGNIFICANCE

"Statistical significance has not been assessed."

### REQUEST LIKE-FOR-LIKE DATA

"Compare relevant product groups on a consistent basis."

---

## Improvement

Can describe an observation or an established effect; specify which.

## Significant

Distinguish business importance from statistical evidence.

## Stable

Needs a defined time-ordered process assessment.

# Phrases you can actually use

## **C** Notice the language / 2 of 2

### SEPARATE LIMITS

"Control limits and specification limits serve different purposes."

### KEEP THE METHOD VISIBLE

"State the subgrouping and monitoring rules."

### AVOID INVENTED DIAGNOSIS

"The signal does not identify its own cause."

### CHECK MEASUREMENT CONSISTENCY

"Confirm the measurement and test basis before combining data."

### ASK FOR TIME ORDER

"Which time-ordered evidence supports the stability claim?"

### AVOID DISMISSING A SIGNAL

"A control signal warrants investigation under the defined procedure."

### PRESERVE USEFUL PROGRESS

"The pilot result is encouraging but limited."

### CLOSE THE FINDING

"Report the counts, observed difference, limitations, and next analysis."

---

### Yield

Attach the test stage, criteria, and denominator.

### In spec

An acceptance statement, not proof of statistical control.

### Caused

Stronger than followed or coincided with.

# Choose the precise message

## **D** Check your understanding

Circle one best answer for each item. Use the stated purpose and case facts, not the option that merely sounds most polite.

### 1. Which announcement is supported?

- A** The process change is proven to improve every product by two relative percent.
- B** The pilot proves the process is statistically stable.
- C** Different denominators make both observed percentages meaningless.
- D** The pilot observed 92% versus 90% yield, with different product mixes and no causal conclusion.

### 2. What is the approximate relative increase from 90% to 92%?

- A** 92.00%
- B** 2.22%
- C** 2.00%
- D** 20.00%

### 3. Which next analysis addresses the product-mix problem?

- A** Compare only the absolute pass counts without the tested populations.
- B** Separate comparable product groups and review consistent test bases.
- C** Pool both lots into one aggregate percentage and omit the product breakdown.
- D** Equalize lot sample sizes while retaining their different product compositions.

### 4. Which statement distinguishes control and specification limits?

- A** Control limits monitor behavior; specification limits define acceptance boundaries.
- B** Control limits are customer acceptance boundaries; specification limits summarize observed variation.
- C** A process within its specification limits necessarily has stable statistical behavior.
- D** A process within its control limits necessarily meets all product requirements.

Answers and explanations: page 91.

# Two percentage points, not a proven process gain

## E Conversations / turns 1-10

04

Complete the numbered gaps. Use each bank entry once, without changing its form. Cover the answers at the end until you finish.

### WORD + PHRASE BANK

percentage point / control signal / specification limit / yield learning / stratification / product mix / measurement system / causal attribution / process stability / pass count

- 01 **Elena:** The pilot reached ninety-two percent against ninety percent in the reference lot. Can we announce that the process change caused a two-percent yield improvement?
- 02 **Ravi:** Begin with the <sup>1</sup> \_\_\_\_\_ and denominator. The pilot passed ninety-two of one hundred tested die; the reference passed nine hundred of one thousand.
- 03 **Elena:** Put both counts beside the percentages. A reader needs to see that one result comes from a hundred die and the other from a thousand.
- 04 **Ravi:** Correct. Also state the <sup>2</sup> \_\_\_\_\_. The lots contain different product compositions, so the two percentages are not a controlled comparison of the process change alone.
- 05 **Elena:** The wording about two percent needs correction too. The absolute difference between ninety and ninety-two percent is two percentage points.
- 06 **Ravi:** Yes, use <sup>3</sup> \_\_\_\_\_ for that absolute change. Relative to the ninety-percent starting value, the increase is approximately two point two two percent.
- 07 **Elena:** I want the team to see the pilot as useful progress without overstating what caused the result. The change happened before the pilot, but timing is not enough.
- 08 **Ravi:** Avoid unsupported <sup>4</sup> \_\_\_\_\_. The result can guide further investigation while the product mix and other relevant differences remain visible.
- 09 **Elena:** Can you break the results down by product and confirm the test basis? I want to see whether the mix is driving the overall difference.
- 10 **Ravi:** Use <sup>5</sup> \_\_\_\_\_ to examine the appropriate groups. It will not automatically prove causality, but it addresses one important weakness in the aggregate comparison.

# The conversation, continued

## E Conversations / turns 11-20

04

Continue with the word bank on page 37.

- 11 **Elena:** The draft also says the pilot demonstrates stable production. We have not presented a time-ordered analysis, only these two lot summaries.
- 12 **Ravi:** Then remove that <sup>6</sup> \_\_\_\_\_ claim. A favorable summary percentage does not establish how the process behaves over time or whether a monitoring rule has signaled a change.
- 13 **Elena:** We have acceptance criteria for individual test results. Some colleagues are treating those limits as though they were the same as the chart's monitoring boundaries.
- 14 **Ravi:** A <sup>7</sup> \_\_\_\_\_ defines an acceptance boundary. A control limit has a different purpose, so passing the specification is not proof that the process is in statistical control.
- 15 **Elena:** If a monitoring rule signals something unusual, the team needs to investigate. But the signal itself does not identify which tool or material caused it.
- 16 **Ravi:** Exactly. A <sup>8</sup> \_\_\_\_\_ is a prompt for the defined investigation process. Do not convert it directly into a root-cause statement or an automatic disposition for every unit.
- 17 **Elena:** We also need to make sure the measurements and test criteria are consistent before combining records. Otherwise the next comparison could carry a different hidden change.
- 18 **Ravi:** Check the <sup>9</sup> \_\_\_\_\_ and test basis. Clear denominators are essential, but they do not solve inconsistencies in how observations were produced or classified.
- 19 **Elena:** The revised update will report the counts, the two-point difference, and the changed product mix. It will leave significance, stability, and causality unclaimed pending the appropriate analysis.
- 20 **Ravi:** That supports <sup>10</sup> \_\_\_\_\_. The pilot remains informative, and the team gets a precise next question instead of a success claim that the present data cannot establish.

### ANSWERS / EXPLANATIONS: PAGE 92

1. pass count | 2. product mix | 3. percentage point | 4. causal attribution | 5. stratification | 6. process stability | 7. specification limit | 8. control signal | 9. measurement system | 10. yield learning

# Percentage points need a denominator

## F Say it

A trial passes 48 of 50 tested die, or 96%. A reference passes 450 of 500, or 90%. Product mixes differ, and no causal analysis is complete.

Complete the four gaps. Use each bank entry once.

### WORD + PHRASE BANK

denominator / product mix / causal attribution / percentage points

Engineer: "Put 48 passing die over the trial's 50-die <sup>1</sup> \_\_\_\_\_."

Reviewer: "Then show the changed <sup>2</sup> \_\_\_\_\_ beside the reference comparison, not in a separate appendix."

Engineer: "I will report the difference as six <sup>3</sup> \_\_\_\_\_, not six relative percent."

Reviewer: "Good. Leave the claim that the process caused it open until we have supported <sup>4</sup> \_\_\_\_\_."

### ANSWERS / EXPLANATIONS: PAGE 92

1. denominator | 2. product mix | 3. percentage points | 4. causal attribution

## Rehearse with a fixed script

1. Check the answers. Read both pass counts and denominators before reading their percentages.
2. Switch roles. Read turns 5-16, emphasizing percentage points, product mix, and the two types of limits.
3. Complete and check the transfer. Read the corrected exchange without turning the observed difference into a causal claim.

### WORKING ALONE

Read both roles, pausing for two seconds when the speaker changes. Repeat with the bank covered, then check any missed expressions.

# Twenty-four detections are not a confirmed source

## A Read the situation

**Describe inspection findings with location, area, timing, and classification while avoiding premature contamination-source claims.**

Inspection engineer Hana and contamination-control engineer Oscar review a note saying cleanroom problem. The underlying record contains 24 particle-like detections over 200 square centimeters inspected on wafers from lot P8 after tool C2 between 9 and 10 a.m. The map shows an edge cluster. No comparable pre-process scan exists, defect review is incomplete, and no source is confirmed. The lot is held under the local procedure. The team must distinguish detections, confirmed defects, added defects, and airborne particle measurements.

PEOPLE / Hana: Inspection engineer | Oscar: Contamination-control engineer

### DESCRIBE THE PATTERN BEFORE NAMING THE SOURCE

A location pattern can focus an investigation without proving where contamination originated. State the observation and its limits, then request traceable process and inspection records. Avoid turning a vague cleanroom label into blame for a team, operator, or tool before the evidence supports it.

## Briefing checks

Circle one answer in each item. Use the briefing.

1. What does the inspection record contain?

**A** 24 particle-like detections over 200 square centimeters

**C** An airborne count per cubic meter

**B** 24 confirmed electrical failures on every wafer

**D** A measured increase of 24 defects from a matched pre-scan

2. What does the map show?

**A** Uniform distribution across every layer

**C** A completed source elimination study

**B** A confirmed cleanroom air source

**D** An edge cluster

3. Why cannot the 24 detections be called 24 added defects?

**A** The lot has an identifier.

**C** No comparable pre-process scan exists, and review is incomplete.

**B** Surface maps always measure airborne particles.

**D** The inspected area is known.

Answers and explanations: [page 93](#).

# The vocabulary that does the work

## B Find the words / 1 of 2

Read the definitions and say the common word combinations aloud.

### inspection detection

A feature flagged by an inspection method for assessment.

**IN USE** review inspection detections

### defect review

Examination used to classify and interpret detected features.

**IN USE** complete defect review

### defect density

Defects per unit area on a stated counting and inspection basis.

**IN USE** report defect density

### inspected area

The surface area included in an inspection result.

**IN USE** state the inspected area

### particle-like signature

An observed appearance consistent with a particle, pending review.

**IN USE** classify a particle-like signature

### nuisance detection

A flagged feature not relevant to the defect concern being assessed.

**IN USE** filter nuisance detections

### killer defect

A defect that causes functional failure in its particular context.

**IN USE** assess killer-defect relevance

### defect classification

Assignment of detected features to defined categories.

**IN USE** refine defect classification

### wafer map

A spatial representation of results on a wafer.

**IN USE** examine the wafer map

### edge cluster

A concentration of observations near a wafer's edge.

**IN USE** investigate an edge cluster

### spatial signature

A location pattern in measured or inspected results.

**IN USE** compare spatial signatures

### added defect

A defect introduced between comparable inspection points.

**IN USE** quantify added defects

### Precision check

The observed detection density is 24 divided by 200, or 0.12 detections per square centimeter. Do not call it a confirmed killer-defect density. Classification is incomplete, and no comparable pre-scan establishes how many detections were introduced by C2.

# The vocabulary that does the work

## B Find the words / 2 of 2

Read the definitions and say the common word combinations aloud.

### pre-process scan

Inspection performed before a specified process step.

**IN USE** preserve the pre-process scan

### post-process scan

Inspection performed after a specified process step.

**IN USE** compare post-process scans

### airborne particle count

A count of particles in a defined air sample or volume.

**IN USE** review airborne particle counts

### surface contamination

Unwanted material present on a surface.

**IN USE** investigate surface contamination

### cross-contamination

Transfer of unwanted material between areas, items, or processes.

**IN USE** prevent cross-contamination

### contamination source

The origin of unwanted material.

**IN USE** identify the contamination source

### particle excursion

An unusual particle-related result under defined monitoring rules.

**IN USE** investigate a particle excursion

### inspection sensitivity

The ability of an inspection method to detect specified features.

**IN USE** confirm inspection sensitivity

### size threshold

The size boundary used for counting or classifying features.

**IN USE** state the size threshold

### exposure history

The record of relevant environments and handling encountered.

**IN USE** reconstruct exposure history

### suspect interval

A defined period being examined for possible affected material.

**IN USE** bound the suspect interval

### containment scope

The material or operations covered by a temporary restriction.

**IN USE** define containment scope

### Precision check

A surface result per square centimeter is not an airborne count per cubic meter. Measurement type, inspected area or air volume, size threshold, and sensitivity belong with the number. Similar timing or spatial patterns guide investigation but do not prove a source.

# Phrases you can actually use

## C Notice the language / 1 of 2

### REPLACE THE VAGUE LABEL

"The record shows particle-like surface detections, not a confirmed cleanroom source."

### NAME THE DENSITY BASIS

"That is 0.12 detections per square centimeter."

### BOUND THE INTERVAL

"These records follow C2 between 9 and 10 a.m."

### AVOID AN ADDED-DEFECT CLAIM

"We lack a comparable pre-process scan."

### GIVE THE COUNT AND AREA

"There were 24 detections over 200 square centimeters inspected."

### LOCATE THE PATTERN

"The wafer map shows an edge cluster."

### PRESERVE THE UNCERTAINTY

"Defect classification remains incomplete."

### SEPARATE MEASUREMENT TYPES

"Surface detections are not airborne particle counts."

---

### Particle-like

An appearance classification, not necessarily a confirmed material identity.

### Added

Needs comparable before-and-after evidence.

### Density

Requires a count definition and area or volume denominator.

# Phrases you can actually use

## C Notice the language / 2 of 2

### ASK ABOUT SENSITIVITY

"Were the inspection settings and size threshold consistent?"

### TRACE THE MATERIAL

"Link the map to the lot, wafers, and process history."

### DEFINE THE NEXT REVIEW

"Review the detections and relevant exposure history."

### ASK ABOUT SCOPE

"Which material is included in the current containment?"

### AVOID PREMATURE BLAME

"The spatial signature does not confirm the source."

### KEEP THE RESTRICTION CLEAR

"P8 remains held under the local procedure."

### AVOID AUTOMATIC YIELD CLAIMS

"Not every flagged feature is a confirmed killer defect."

### CLOSE THE NOTE

"Report observations, limits, containment status, and investigation ownership."

---

### Source

A causal conclusion, not merely the last tool visited.

### Cluster

A spatial pattern that can guide but not finish an investigation.

### Cleanroom problem

Too broad to locate the evidence or assign a justified cause.

# Choose the precise message

## **D** Check your understanding

Circle one best answer for each item. Use the stated purpose and case facts, not the option that merely sounds most polite.

### 1. What is the observed detection density?

- A** 0.12 particles per cubic meter of air
- B** 0.12 detections per square centimeter
- C** 12 detections per square centimeter
- D** 8.33 detections per square centimeter

### 2. Which conclusion is unsupported?

- A** The records are after C2.
- B** The map shows an edge cluster.
- C** Defect review is incomplete.
- D** C2 introduced all 24 detections.

### 3. Which next record is useful for the investigation?

- A** A whole-fab monthly total used to assign the source for this lot
- B** A surface map converted to an airborne count without an air sample
- C** Traceable wafer, inspection-setting, timing, and exposure information
- D** A count from another layer using a different threshold, treated as a matched pre-scan

### 4. Which distinction is correct?

- A** Surface detection density and airborne particle counts use different measurement bases.
- B** A post-process detection count equals an added-defect count without a pre-scan.
- C** An inspection's flagged-feature count is already the confirmed killer-defect count.
- D** An edge pattern identifies the contamination source without further review.

Answers and explanations: [page 93](#).

# Twenty-four detections are not a confirmed source

## E Conversations / turns 1-10

05

Complete the numbered gaps. Use each bank entry once, without changing its form. Cover the answers at the end until you finish.

### WORD + PHRASE BANK

exposure history / airborne particle count / edge cluster / contamination source / Defect review / containment scope / inspected area / pre-process scan / added defect / inspection sensitivity

- 01 **Hana:** The review note says cleanroom problem, but that label is much broader than the evidence. We have twenty-four particle-like detections on the inspected surfaces from P8.
- 02 **Oscar:** Start with the <sup>1</sup> \_\_\_\_\_. The record covers two hundred square centimeters, so the number can be interpreted on a defined surface basis rather than as an unexplained total.
- 03 **Hana:** That gives zero point one two detections per square centimeter. I should keep the word detections because the detailed classification has not been completed.
- 04 **Oscar:** Correct. <sup>2</sup> \_\_\_\_\_ is still open. A flagged feature is not automatically a confirmed defect of the relevant type, much less a demonstrated electrical failure.
- 05 **Hana:** The map clusters near the edge. I will send the wafer IDs and coordinates so the team can examine the pattern without guessing its source.
- 06 **Oscar:** Call it an <sup>3</sup> \_\_\_\_\_. Describe where it appears and connect the map to the wafer IDs, rather than turn the pattern into an immediate diagnosis of the cleanroom.
- 07 **Hana:** The records follow C2 between nine and ten a.m. Someone has rewritten that as C2 added twenty-four particles, although we do not have a comparable earlier scan.
- 08 **Oscar:** We cannot claim an <sup>4</sup> \_\_\_\_\_ count on that basis. After C2 establishes the recorded sequence, not proof that C2 introduced every detected feature.
- 09 **Hana:** A matched earlier inspection would help distinguish what was already present from what appeared later. We would also need to check the inspection basis before comparing counts.
- 10 **Oscar:** Exactly. A <sup>5</sup> \_\_\_\_\_ must be comparable with the later scan. Different sensitivity, locations, or thresholds could otherwise make a count difference look like process-added contamination.

# The conversation, continued

## E Conversations / turns 11-20

Continue with the word bank on page 46.

- 11 **Hana:** Facilities sent an air-monitoring result too. I will keep it separate from these surface counts; the units and measurement methods are different.
- 12 **Oscar:** An <sup>6</sup> \_\_\_\_\_ is a different measurement. A value per air volume cannot be treated as the same quantity as detections per inspected surface area.
- 13 **Hana:** I will include the inspection recipe and threshold with the maps. That will let reviewers see whether the method was capable of detecting the features being discussed.
- 14 **Oscar:** Keep <sup>7</sup> \_\_\_\_\_ visible. A tool finding more features after a settings change does not by itself prove that the process produced more contamination.
- 15 **Hana:** P8 remains on hold under the local procedure. The investigation team needs to say which material and time interval are included, rather than leave the restriction implicit.
- 16 **Oscar:** Define the <sup>8</sup> \_\_\_\_\_ through the authorized process. A precise scope supports coordination without turning this discussion into permission to release or extend the hold arbitrarily.
- 17 **Hana:** We can reconstruct the lot's recent handling and process history alongside the spatial pattern. That should help distinguish possible explanations and identify useful follow-up evidence.
- 18 **Oscar:** Review the <sup>9</sup> \_\_\_\_\_. It can guide source investigation, but a shared tool or time interval is not enough on its own to establish the origin.
- 19 **Hana:** The corrected note will give the count, area, edge pattern, interval, and incomplete classification. It will state that introduction by C2 and an airborne source are unconfirmed.
- 20 **Oscar:** That keeps the <sup>10</sup> \_\_\_\_\_ question open on an accurate basis. We can investigate efficiently while preserving the distinction between an observation, a hypothesis, and a confirmed cause.

### ANSWERS / EXPLANATIONS: PAGE 94

1. inspected area | 2. Defect review | 3. edge cluster | 4. added defect | 5. pre-process scan | 6. airborne particle count | 7. inspection sensitivity | 8. containment scope | 9. exposure history | 10. contamination source

# Count the detections, not an invented cause

## F Say it

A post-process scan flags 12 features over 100 square centimeters. Classification is pending, and no comparable pre-process scan exists. No contamination source is confirmed.

Complete the four gaps. Use each bank entry once.

### WORD + PHRASE BANK

defect review / pre-process scan / contamination source / inspected area

Engineer: "Keep 100 square centimeters beside the count as the

<sup>1</sup> \_\_\_\_\_."

Reviewer: "And leave the twelve flagged features unclassified pending

<sup>2</sup> \_\_\_\_\_."

Engineer: "I will request a comparable <sup>3</sup> \_\_\_\_\_ before reporting any process-added count."

Reviewer: "Thank you. Do not name the last tool as the <sup>4</sup> \_\_\_\_\_ without supporting investigation."

### ANSWERS / EXPLANATIONS: PAGE 94

1. inspected area | 2. defect review | 3. pre-process scan | 4. contamination source

## Rehearse with a fixed script

1. Check the answers. Read 24 detections over 200 square centimeters and the resulting density aloud.
2. Switch roles for turns 5-14. Keep the edge pattern, possible source, and surface-versus-air distinction explicit.
3. Complete and check the transfer. Read the evidence requests without naming an unconfirmed contamination source.

### WORKING ALONE

Read both roles, pausing for two seconds when the speaker changes. Repeat with the bank covered, then check any missed expressions.

# Same recipe, different means

## A Read the situation

**Distinguish recipe identity, specification performance, tool matching, and equipment availability in a transfer discussion.**

Equipment engineer Victor and process engineer Asha compare tools A and B using the same identified R7 recipe and an agreed comparable measurement basis. Their film-thickness sample means are 98 nm and 102 nm. The local mean specification is 95 to 105 nm, but the tool-matching criterion requires the means to differ by no more than 2 nm. Tool B was available for 900 of a defined 1,000-minute window. A colleague uses the identical recipe and 90% availability to claim B is an approved interchangeable backup. No transfer approval is recorded.

PEOPLE / Victor: Equipment engineer | Asha: Process engineer

06

### NAME WHICH READINESS QUESTION YOU ANSWERED

Available, in specification, matched, and released describe different conditions. A tool may satisfy one and not another. Give each claim its own evidence and criterion so production planning does not convert equipment uptime into process-transfer approval.

## Briefing checks

Circle one answer in each item. Use the briefing.

1. How far apart are the sample means?

A 2 nm

B 7 nm

C 200 nm

D 4 nm

2. Do the means meet the stated matching criterion?

A Yes; both tools use R7.

B The availability percentage automatically settles matching.

C No; 4 nm exceeds the allowed 2 nm difference.

D Yes; both means are within the mean specification.

3. What does 90% describe here?

A Availability for 900 of the defined 1,000 minutes

B The proportion of wafers passing final test

C A 90% probability that the tools are interchangeable

D A recorded production transfer approval

Answers and explanations: page 95.

# The vocabulary that does the work

## B Find the words / 1 of 2

Read the definitions and say the common word combinations aloud.

### tool matching

Demonstrating comparable tool performance against defined criteria.

**IN USE assess tool matching**

### chamber matching

Demonstrating comparable performance between process chambers.

**IN USE verify chamber matching**

### recipe revision

The identified version of a process recipe.

**IN USE confirm the recipe revision**

### setpoint

A commanded target value for a controlled variable.

**IN USE record the setpoint**

### actual value

The measured or achieved value rather than the commanded target.

**IN USE compare setpoint and actual value**

### equipment state

The recorded operational condition of a tool.

**IN USE report equipment state**

### availability

The share of a defined time basis during which equipment is available.

**IN USE calculate equipment availability**

### utilization

The share of a stated time basis during which equipment is used.

**IN USE state the utilization denominator**

### scheduled downtime

Planned time when equipment is unavailable for the defined use.

**IN USE plan scheduled downtime**

### unscheduled downtime

Unplanned time when equipment is unavailable.

**IN USE investigate unscheduled downtime**

### preventive maintenance

Planned work intended to reduce equipment failures or deterioration.

**IN USE schedule preventive maintenance**

### corrective maintenance

Work to restore equipment after a fault or failure.

**IN USE complete corrective maintenance**

### Precision check

Both sample means meet the local 95-to-105 nm mean specification. Their difference is 4 nm, so they fail the separate 2 nm matching criterion. This does not establish the status of every individual wafer, whose measurements are not supplied.

# The vocabulary that does the work

## B Find the words / 2 of 2

Read the definitions and say the common word combinations aloud.

### MTBF

Mean time between failures under a stated operating and event definition.

**IN USE report MTBF**

### MTTR

Mean time to repair or restore under the stated local definition.

**IN USE define the MTTR basis**

### chamber condition

The physical and operating state of a process chamber.

**IN USE assess chamber condition**

### seasoning

Controlled processing used to establish a required chamber condition.

**IN USE verify seasoning status**

### calibration status

The current documented state of required measurement calibration.

**IN USE check calibration status**

### qualification wafer

A wafer used to assess a specified tool or process condition.

**IN USE evaluate a qualification wafer**

### monitor wafer

A wafer used to track selected process or tool behavior.

**IN USE review monitor-wafer results**

### matching criterion

A defined limit for acceptable performance difference.

**IN USE apply the matching criterion**

### process transfer

Moving a defined process to another tool or location.

**IN USE approve process transfer**

### backup tool

An alternative tool intended for use under defined conditions.

**IN USE qualify a backup tool**

### release status

Whether the required authorization for a specified use is in place.

**IN USE confirm release status**

### offset correction

A controlled adjustment addressing a measured systematic difference.

**IN USE assess offset correction**

### Precision check

Availability is 900 divided by 1,000, or 90%, on this stated basis. It does not measure process matching, output yield, or production-use approval. Availability and utilization definitions vary, so keep the time denominator explicit.

# Phrases you can actually use

## C Notice the language / 1 of 2

### CONFIRM THE RECIPE

"Both tools used the identified R7 revision."

### STATE THE MEANS

"A measured 98 nm and B measured 102 nm."

### APPLY THE RIGHT CRITERION

"Matching allows no more than a 2 nm difference."

### BOUND THE EVIDENCE

"These are sample means, not every individual wafer result."

### SEPARATE INPUTS FROM OUTPUTS

"Identical setpoints do not prove identical performance."

### GIVE THE DIFFERENCE

"The means are 4 nm apart."

### SEPARATE THE SPECIFICATIONS

"Both means are in spec, but the pair is not matched."

### ASK ABOUT CHAMBER STATE

"Were the relevant chamber conditions documented?"

---

### Same

Identify whether recipe, settings, product, or measured outcome is the same.

### Available

Describes a time or operational state, not qualification.

### In spec

Specify the metric; a passing mean does not prove every unit passed.

# Phrases you can actually use

## C Notice the language / 2 of 2

### CHECK MEASUREMENT STATUS

"Confirm the calibration and measurement basis."

### KEEP THE PERCENTAGE LIMITED

"That 90% figure is availability, not yield."

### ROUTE THE INVESTIGATION

"Review the observed offset before proposing a correction."

### CLARIFY THE TRANSFER

"Which process, product, and tool combination is proposed?"

### STATE AVAILABILITY

"B was available for 900 of the defined 1,000 minutes."

### AVOID AUTOMATIC SUBSTITUTION

"Backup status needs the required qualification and approval."

### PRESERVE AUTHORIZATION

"Do not change the recipe from this comparison alone."

### CLOSE THE STATUS

"Report specification, matching, availability, and release separately."

06

### Matched

Requires an actual matching criterion and supporting evidence.

### Backup

A planning role that does not automatically authorize production use.

### Offset

A measured difference to investigate, not an instruction to adjust settings.

# Choose the precise message

## D Check your understanding

Circle one best answer for each item. Use the stated purpose and case facts, not the option that merely sounds most polite.

### 1. Which status report is accurate?

- A Tool B is approved because its availability is 90%.
- B Every wafer from both tools is proven in specification.
- C Both means meet the mean specification; their 4 nm difference fails the 2 nm matching limit.
- D Both tools are matched because 98 and 102 lie within 95 to 105.

### 2. What does using the same recipe establish?

- A A matching recipe revision substitutes for product-specific transfer approval.
- B The identified recipe input is shared, not that output performance is interchangeable.
- C The two tools have equivalent physical chamber conditions throughout the run.
- D The same nominal setpoints remove the need to compare actual results.

### 3. Which calculation matches the availability basis?

- A 100 unavailable minutes divided by 1,000 gives 10% availability.
- B The available-time count alone establishes utilization without a used-time count.
- C 900 divided by 1,000 equals 90%.
- D 900 divided by the 100 unavailable minutes equals the availability fraction.

### 4. What remains necessary before calling B an approved interchangeable backup?

- A Only a passing sample mean against the broad mean specification
- B The required matching or qualification evidence and transfer approval
- C Only a repeated statement that R7 is installed
- D Only a higher availability percentage

Answers and explanations: page 95.

# Same recipe, different means

## E Conversations / turns 1-10

Complete the numbered gaps. Use each bank entry once, without changing its form. Cover the answers at the end until you finish.

### WORD + PHRASE BANK

matching criterion / Tool matching / release status / chamber condition / calibration status / availability / equipment state / process transfer / actual value / recipe revision

06

- 01 **Victor:** Tool B has the same R7 recipe as A and was available for ninety percent of the window. The planning note calls it an interchangeable backup.
- 02 **Asha:** That combines different questions. <sup>1</sup> \_\_\_\_\_ depends on the measured outcomes and our criterion, not only the recipe name or the amount of time the tool was available.
- 03 **Victor:** We confirmed the same recipe revision and an agreed measurement basis. A's sample mean was ninety-eight nanometers, while B's was one hundred two.
- 04 **Asha:** The <sup>2</sup> \_\_\_\_\_ allows a difference of no more than two nanometers. These means differ by four, so this comparison does not establish an acceptable match.
- 05 **Victor:** Both means are within the local mean specification of ninety-five to one hundred five. That explains why someone thought the comparison had passed.
- 06 **Asha:** The <sup>3</sup> \_\_\_\_\_ has to be assessed against the correct question. Each mean passes that specification, but the pair fails the separate matching limit.
- 07 **Victor:** I will say both sample means meet that limit. We have not reviewed every individual measurement, so I cannot extend that claim to every wafer.
- 08 **Asha:** Agreed. Confirm the <sup>4</sup> \_\_\_\_\_ and report the evidence at its actual level. Do not let a summary mean or a shared input become a claim about every processed unit.
- 09 **Victor:** The observed difference needs investigation. Chamber history and maintenance condition may be relevant, but neither has been established as the cause from these numbers.
- 10 **Asha:** Review <sup>5</sup> \_\_\_\_\_ and the other appropriate records. Identical commanded settings do not guarantee identical physical conditions or measurement behavior.

# The conversation, continued

## E Conversations / turns 11-20

Continue with the word bank on page 55.

- 11 **Victor:** I will bring the calibration and monitor records. Can we review those before anyone suggests changing R7 to remove the offset?
- 12 **Asha:** Check <sup>6</sup> \_\_\_\_\_ through the approved process. We should not prescribe a recipe adjustment before the team understands the relevant evidence and authorization requirements.
- 13 **Victor:** For the uptime figure, B was available for nine hundred minutes out of a defined one-thousand-minute window. That calculation gives ninety percent.
- 14 **Asha:** Label it <sup>7</sup> \_\_\_\_\_ and retain the denominator. It does not mean ninety percent yield or ninety percent confidence that the tool can substitute for A.
- 15 **Victor:** We also have no usage-time figure here, so we should not relabel availability as utilization. The two measures can use different time bases and answer different questions.
- 16 **Asha:** Correct. The <sup>8</sup> \_\_\_\_\_ record helps explain the time measure, but an available tool is not automatically qualified for this product and process combination.
- 17 **Victor:** The backup proposal remains useful because capacity is constrained, but it needs the appropriate technical assessment and approval before production relies on it.
- 18 **Asha:** Treat it as a <sup>9</sup> \_\_\_\_\_ decision. The defined scope, matching evidence, and required authorization must be clear instead of being inferred from the existence of an alternative tool.
- 19 **Victor:** I will revise the planning note to separate the passing means, failed matching comparison, ninety-percent availability, and missing approval. It will no longer call B interchangeable.
- 20 **Asha:** That preserves the <sup>10</sup> \_\_\_\_\_ accurately. We can pursue the backup option while making its unresolved requirements visible to the people planning production.

06

### ANSWERS / EXPLANATIONS: PAGE 96

1. Tool matching | 2. matching criterion | 3. actual value | 4. recipe revision | 5. chamber condition | 6. calibration status | 7. availability | 8. equipment state | 9. process transfer | 10. release status

# Two different limits

## F Say it

Two tools have sample means of 50 and 53 nm. Both meet a mean specification of 48 to 55 nm. Matching allows at most 1 nm difference. Transfer approval is pending.

Complete the four gaps. Use each bank entry once.

06

### WORD + PHRASE BANK

matching criterion / process transfer / release status / specification

Engineer: "Both means meet the 48-to-55 nm <sup>1</sup> \_\_\_\_\_."

Reviewer: "But their three-nanometer gap misses our one-nanometer <sup>2</sup> \_\_\_\_\_."

Engineer: "Then moving this process to the other tool remains a proposed <sup>3</sup> \_\_\_\_\_."

Reviewer: "Correct. Keep approval pending in the <sup>4</sup> \_\_\_\_\_ field."

### ANSWERS / EXPLANATIONS: PAGE 96

1. specification | 2. matching criterion | 3. process transfer | 4. release status

## Rehearse with a fixed script

1. Check the answers. Read the 98 nm and 102 nm means, then the separate specification and matching limits.
2. Switch roles. Read the availability exchange without replacing availability with yield or utilization.
3. Complete and check the transfer. Read each claim with its own criterion and keep approval pending.

### WORKING ALONE

Read both roles, pausing for two seconds when the speaker changes. Repeat with the bank covered, then check any missed expressions.

# One completed test is not complete qualification

## A Read the situation

**Explain qualification status test by test, preserve configuration scope, and correct an overbroad customer claim.**

Product engineer Diego and reliability engineer Mei review package Q7. Its approved fictional qualification plan requires temperature cycling, high-temperature operating life testing, a highly accelerated stress test, and final review of the evidence. Temperature cycling has passed under its protocol. Operating-life testing is running, and the accelerated stress test has not started. No final approval or accepted substitution is recorded. A customer slide nevertheless says qualification complete. The team must replace that headline with a precise status without describing open tests as failures.

PEOPLE / Diego: Product engineer | Mei: Reliability engineer

07

### GIVE PROGRESS WITHOUT COLLAPSING THE PLAN

A customer may prefer a short headline, but complete must refer to the required plan and configuration. Name the finished test, the open work, and the pending decision. Do not imply that a test still running has passed, or that a test not yet started has failed.

## Briefing checks

Circle one answer in each item. Use the briefing.

1. Which test has passed?

A Temperature cycling

C The highly accelerated stress test

B High-temperature operating life

D Every required test and final review

2. What is the operating-life test status?

A Running

C Not required by the plan

B Passed and approved

D Replaced by an accepted substitution

3. Why is qualification complete unsupported?

A Every package test is optional.

C The package identifier proves approval.

B A running test is automatically a failure.

D Two required tests and final review remain open.

Answers and explanations: [page 97](#).

# The vocabulary that does the work

## **B** Find the words / 1 of 2

Read the definitions and say the common word combinations aloud.

### package substrate

The package structure carrying connections between die and external interfaces.

**IN USE** specify the package substrate

### wire bonding

Forming electrical connections using fine wires.

**IN USE** qualify wire bonding

### flip-chip attachment

Connecting a die face-down through conductive bumps or pillars.

**IN USE** evaluate flip-chip attachment

### underfill

Material placed around interconnections to support a package assembly.

**IN USE** assess underfill integrity

### mold compound

Encapsulating material used in many semiconductor packages.

**IN USE** qualify the mold compound

### solder joint

A soldered electrical and mechanical connection.

**IN USE** inspect solder joints

### BGA

Ball grid array, a package connection format using an array of solder balls.

**IN USE** assess BGA assembly

### QFN

Quad flat no-lead, a package format with exposed connection lands.

**IN USE** review the QFN package

### SiP

System in package, integrating multiple functions or components in one package.

**IN USE** evaluate a SiP design

### TSV

Through-silicon via, a conductive connection passing through silicon.

**IN USE** characterize TSV connections

### package construction

The materials and structural arrangement of a package.

**IN USE** identify package construction

### moisture sensitivity

Susceptibility to moisture-related effects under stated handling or stress conditions.

**IN USE** assess moisture sensitivity

### Precision check

This case uses a specific fictional plan, not a universal mandatory three-test list. One passing test supports its defined result. It does not complete the two open tests or the final review, and it does not establish every application-specific reliability claim.

# The vocabulary that does the work

## B Find the words / 2 of 2

Read the definitions and say the common word combinations aloud.

### delamination

Separation between bonded material layers.

**IN USE inspect for delamination**

### thermal cycling

Repeated temperature changes under a specified test profile.

**IN USE complete thermal cycling**

### HTOL

High-temperature operating life, an operating stress test under defined conditions.

**IN USE report HTOL status**

### HAST

Highly accelerated stress test, a defined accelerated humidity-related stress test.

**IN USE schedule HAST**

### preconditioning

Defined preparation before a test to represent relevant handling or assembly stresses.

**IN USE document preconditioning**

### qualification plan

The defined tests, evidence, scope, and approval requirements for qualification.

**IN USE follow the qualification plan**

### qualification vehicle

The selected device or structure used to support qualification evidence.

**IN USE identify the qualification vehicle**

### test coverage

The requirements or failure concerns addressed by the performed tests.

**IN USE assess qualification test coverage**

### failure analysis

Investigation of failed samples and their causes.

**IN USE initiate failure analysis**

### qualification by similarity

Use of justified related evidence under an accepted qualification approach.

**IN USE assess qualification by similarity**

### final test

Electrical testing at the defined final product stage.

**IN USE complete final test**

### qualification approval

Authorized acceptance of the required qualification evidence.

**IN USE obtain qualification approval**

### Precision check

Related package evidence may sometimes support qualification through an accepted similarity rationale. No such substitution is accepted here. Similar appearance, a shared package name, or a convenient schedule cannot silently change the approved evidence requirements.

# Phrases you can actually use

## **C** Notice the language / 1 of 2

### NAME THE COMPLETED RESULT

"Q7 passed temperature cycling under the stated protocol."

### STATE UNSTARTED WORK

"HAST has not started."

### CORRECT THE HEADLINE

"Qualification is in progress, not complete."

### PRESERVE CONFIGURATION

"Which package construction does the evidence cover?"

### STATE ONGOING WORK

"HTOL is still running."

### KEEP THE PLAN VISIBLE

"The approved plan requires all three tests and final review."

### AVOID TREATING PENDING AS FAILED

"No result is available for the unstarted test."

### ASK ABOUT THE VEHICLE

"Is this the approved qualification vehicle?"

---

## Qualified

Specify product, package, construction, scope, and approval.

## Passed

Applies to a defined test result, not automatically the entire plan.

## Running

A current activity status, not a completed result.

# Phrases you can actually use

## **C** Notice the language / 2 of 2

### CHECK PREPARATION

"Are the required preconditioning records included?"

### QUALIFY RELATED EVIDENCE

"Similarity needs an accepted technical rationale."

### STATE THE APPROVAL GAP

"Final qualification approval is not recorded."

### COMMIT TO THE NEXT UPDATE

"The next report will distinguish completed, running, and unstarted tests."

### SEPARATE PRODUCTION TEST

"Final test is not a substitute for every reliability stress."

### AVOID A SCHEDULE SHORTCUT

"A desired customer date does not remove a required test."

### DESCRIBE COVERAGE

"Identify what each completed test actually addresses."

### CLOSE ACCURATELY

"Keep release decisions with the authorized review process."

---

### Not started

Does not mean failed or waived.

### Similar

Needs a justified and accepted relevance argument.

### Complete

Includes the required evidence review, not only laboratory scheduling.

# Choose the precise message

## **D** Check your understanding

Circle one best answer for each item. Use the stated purpose and case facts, not the option that merely sounds most polite.

### 1. Which customer headline is supported?

- A** Q7 is fully qualified because temperature cycling passed.
- B** Q7 failed HAST because HAST has not started.
- C** Q7 is approved by similarity despite no accepted substitution.
- D** Qualification in progress: temperature cycling passed; HTOL running; HAST not started.

### 2. What does the passing temperature-cycle result establish?

- A** Completion of the operating-life test because both involve temperature
- B** Completion of HAST because both are accelerated stress tests
- C** Qualification of every package variant sharing the Q7 marketing name
- D** The stated test result under its protocol and scope

### 3. When could related evidence replace a required test in this case?

- A** When the related package has the same outside dimensions, without a construction review
- B** When the older report is newer than the current plan, without an applicability assessment
- C** When final electrical test passes, without assessing the reliability evidence gap
- D** Only through the applicable accepted change or similarity process, which has not occurred

### 4. What should final review examine?

- A** Required evidence, scope, configuration, open items, and approval conditions
- B** Only whether the slide uses the word complete
- C** Only the date the first test started
- D** A presumed pass for all tests without final results

[Answers and explanations: page 97.](#)

# One completed test is not complete qualification

## E Conversations / turns 1-10

Complete the numbered gaps. Use each bank entry once, without changing its form. Cover the answers at the end until you finish.

### WORD + PHRASE BANK

thermal cycling / preconditioning / qualification plan / Test coverage / qualification approval / HAST / HTOL / Qualification by similarity / qualification vehicle / package construction

07

- 01 **Diego:** The customer slide says Q7 qualification is complete because temperature cycling passed. We still have two other tests in the plan, so I need a more accurate headline.
- 02 **Mei:** Start from the <sup>1</sup> \_\_\_\_\_. Our approved plan requires all three tests and final evidence review. We have completed one part, not the entire qualification decision.
- 03 **Diego:** The completed test followed its protocol and passed. I want to preserve that positive result rather than make the update sound as though nothing has been achieved.
- 04 **Mei:** Report the <sup>2</sup> \_\_\_\_\_ result clearly. Its value is not diminished by stating the other work accurately, and it should not be expanded beyond its protocol and scope.
- 05 **Diego:** The operating-life test is still running. The lab has not supplied its final result, so the customer should not read the current status as a pass.
- 06 **Mei:** Label <sup>3</sup> \_\_\_\_\_ as running. If a progress checkpoint is available, identify it as interim information without treating it as the completed test outcome.
- 07 **Diego:** The accelerated humidity-related stress test has not started. That is a schedule and evidence gap, but it is not a failed result.
- 08 **Mei:** Exactly. <sup>4</sup> \_\_\_\_\_ remains unstated in this plan. Use that status rather than either a reassuring green result or an unsupported failure label.
- 09 **Diego:** Which package construction should I name on the slide? Q7 alone may cover variants with different materials or assembly details.
- 10 **Mei:** Keep the <sup>5</sup> \_\_\_\_\_ linked to the evidence. Qualification scope should not silently expand to a different variant simply because the marketing name is similar.

# The conversation, continued

## E Conversations / turns 11-20

Continue with the word bank on page 64.

- 11 **Diego:** We should confirm the tested device or structure as well. Otherwise reviewers may not know how the selected samples relate to the configuration being discussed.
- 12 **Mei:** Identify the <sup>6</sup> \_\_\_\_\_ and its approved rationale. Traceability makes it possible to review applicability rather than assume that any available sample represents every product.
- 13 **Diego:** The assembly preparation records also belong in the package. I do not want the final review to discover that important preparation details were omitted from the summary.
- 14 **Mei:** Include the required <sup>7</sup> \_\_\_\_\_ records where the plan calls for them. A passing result needs the relevant preparation and test context to be interpreted correctly.
- 15 **Diego:** Sales wants to reuse results from an older package. I will refer that request for review; we have no accepted substitution for the open tests.
- 16 **Mei:** <sup>8</sup> \_\_\_\_\_ needs an accepted basis under the applicable process. A related package or a desired delivery date does not automatically change the approved requirements.
- 17 **Diego:** We also have final electrical test records, but those should not be presented as though they address every stress-related concern in the qualification plan.
- 18 **Mei:** Correct. <sup>9</sup> \_\_\_\_\_ differs between production checks and reliability assessments. Each result needs to be connected to what it actually evaluates, not treated as interchangeable evidence.
- 19 **Diego:** I will change the headline to qualification in progress, list the three statuses, and keep the configuration and outstanding review visible. We will not announce release from this slide.
- 20 **Mei:** That preserves <sup>10</sup> \_\_\_\_\_ as a separate decision. The next update can show genuine progress while keeping the unresolved tests and final review clear.

07

### ANSWERS / EXPLANATIONS: PAGE 98

1. qualification plan | 2. thermal cycling | 3. HTOL | 4. HAST | 5. package construction | 6. qualification vehicle | 7. preconditioning | 8. Qualification by similarity | 9. Test coverage | 10. qualification approval

# Pending is neither passed nor failed

## F Say it

A fictional package plan requires tests A, B, and C plus final review. A passed, B is running, and C has not started. No substitution or final approval is recorded.

Complete the four gaps. Use each bank entry once.

07

### WORD + PHRASE BANK

approval / plan / evidence / rationale

Engineer: "Keep A's passing result as test <sup>1</sup> \_\_\_\_\_, not a claim that every requirement is complete."

Reviewer: "B and C are still required by the approved <sup>2</sup> \_\_\_\_\_."

Engineer: "We would need an accepted technical <sup>3</sup> \_\_\_\_\_ before substituting related results."

Reviewer: "Until the tests and final review are complete, do not announce qualification <sup>4</sup> \_\_\_\_\_."

### ANSWERS / EXPLANATIONS: PAGE 98

1. evidence | 2. plan | 3. rationale | 4. approval

## Rehearse with a fixed script

1. Check the answers. Read the three test statuses, expanding HTOL and HAST using the vocabulary pages.
2. Switch roles for turns 9-20. Keep construction, preparation, related evidence, and final approval distinct.
3. Complete and check the transfer. Read the pending work without changing it into either a pass or a failure.

### WORKING ALONE

Read both roles, pausing for two seconds when the speaker changes. Repeat with the bank covered, then check any missed expressions.

# A possible slot is not tape-out readiness

## A Read the situation

**Negotiate a requested date by separating design sign-off, data delivery, foundry acceptance, and capacity confirmation.**

Customer design lead Sofia asks foundry liaison Ken to move a planned October 15 tape-out to October 8. Four design-rule-check findings and one layout-versus-schematic mismatch remain unresolved in the agreed fictional PDK revision P3. The foundry has mentioned a possible earlier slot but has not reserved or accepted it. No complete sign-off package is available, and no waiver is approved. Ken must communicate the blockers and conditional next steps without promising a wafer start, finished samples, or shipment.

PEOPLE / Sofia: Customer design lead | Ken: Foundry liaison

08

### SEPARATE THE MILESTONES UNDER DATE PRESSURE

Tape-out, foundry data acceptance, wafer start, and finished samples are related but distinct milestones. State which date is requested and which prerequisites remain open. A possible capacity opportunity should trigger a feasibility discussion, not erase design checks or become a shipment promise.

## Briefing checks

Circle one answer in each item. Use the briefing.

1. Which earlier date is requested?

A October 15 completed wafer sort

B October 8 tape-out

C October 8 finished samples

D October 15 shipment

2. What remains unresolved in the design checks?

A An approved waiver for all findings

B Four DRC findings and one LVS mismatch

C A completed sign-off package with no open items

D Only a customer shipping address

3. What is the capacity status?

A A confirmed wafer start

B A guaranteed sample delivery

C An approved bypass of design sign-off

D A possible earlier slot, not reserved or accepted

Answers and explanations: page 99.

# The vocabulary that does the work

## B Find the words / 1 of 2

Read the definitions and say the common word combinations aloud.

### foundry

A manufacturer producing semiconductor devices for other companies.

**IN USE** coordinate with the foundry

### fabless company

A chip-design company that contracts out fabrication.

**IN USE** support a fabless customer

### IDM

Integrated device manufacturer, combining chip design and manufacturing.

**IN USE** work with an IDM

### PDK

Process design kit: process-specific models, rules, and design resources.

**IN USE** confirm the PDK revision

### design rule

A process-specific constraint governing layout features or relationships.

**IN USE** check design rules

### DRC

Design rule checking, testing layout against applicable rule definitions.

**IN USE** resolve DRC findings

### LVS

Layout versus schematic checking, comparing extracted layout connectivity with the schematic.

**IN USE** close an LVS mismatch

### rule deck

The implemented rules used by a physical-verification tool.

**IN USE** identify the rule deck

### sign-off

The required formal completion and approval of defined checks.

**IN USE** complete design sign-off

### tape-out

Release of the approved design dataset for the intended fabrication handoff.

**IN USE** plan the tape-out

### GDSII

A widely used layout-data format for integrated circuit design.

**IN USE** deliver a GDSII dataset

### OASIS

A compact integrated-circuit layout interchange format.

**IN USE** verify the OASIS file

### Precision check

October 8 is a requested tape-out date, not an accepted slot or a finished-sample date. The open DRC and LVS items remain blockers to the stated sign-off package. The case does not provide an approved waiver or permission to bypass checks.

# The vocabulary that does the work

## B Find the words / 2 of 2

Read the definitions and say the common word combinations aloud.

### mask data preparation

Processing layout data for mask manufacture.

**IN USE** schedule mask data preparation

### mask set

The collection of masks needed for a defined fabrication flow.

**IN USE** release the mask set

### engineering change order

A controlled record authorizing a defined engineering change, abbreviated ECO.

**IN USE** approve an engineering change order

### waiver

An authorized exception to a specific requirement under an applicable process.

**IN USE** request a defined waiver

### data acceptance

Confirmation that a delivered dataset meets the receiving process's requirements.

**IN USE** obtain foundry data acceptance

### capacity reservation

A confirmed allocation of manufacturing capacity.

**IN USE** confirm capacity reservation

### wafer start

Entry of a wafer lot into the defined fabrication flow.

**IN USE** confirm the wafer start

### cycle time

Elapsed time through a defined manufacturing flow or stage.

**IN USE** state fabrication cycle time

### engineering lot

A lot used for development or engineering evaluation.

**IN USE** plan an engineering lot

### risk production

Production begun before all defined qualification or maturity conditions are complete.

**IN USE** assess risk-production approval

### respin

A revised design iteration requiring another fabrication cycle.

**IN USE** evaluate the need for a respin

### delivery commitment

An agreed promise about a specified deliverable and date.

**IN USE** qualify the delivery commitment

### Precision check

PDK and rule-deck versions affect what physical verification means. Keep results connected to the agreed revision. A clean check run also has a defined scope; it does not guarantee first-silicon performance, complete qualification, or a confirmed manufacturing schedule.

# Phrases you can actually use

## **C** Notice the language / 1 of 2

### NAME THE REQUESTED MILESTONE

"You are requesting October 8 tape-out."

### STATE THE OPEN CHECKS

"Four DRC findings and one LVS mismatch remain open."

### AVOID SILENT VERSION CHANGES

"Keep the rule deck and design dataset identified."

### ASK ABOUT ACCEPTANCE

"What is required for the foundry to accept the dataset?"

### SEPARATE LATER DATES

"That is not a wafer-start or sample-delivery commitment."

### CONFIRM THE BASIS

"The checks use the agreed PDK revision P3."

### QUALIFY THE SLOT

"The foundry mentioned a possibility, not a reservation."

### KEEP SIGN-OFF VISIBLE

"The complete sign-off package is not yet available."

08

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## Slot

Clarify whether possible, held, reserved, or confirmed.

## Ready

Identify which handoff and its required checks.

## Tape-out

Not interchangeable with wafer start or finished-device delivery.

# Phrases you can actually use

## **C** Notice the language / 2 of 2

### **BOUND THE EXCEPTION**

"No waiver is approved for these findings."

### **REQUEST A FEASIBLE SEQUENCE**

"Align check closure, data review, and capacity confirmation."

### **CLARIFY OWNERSHIP**

"Who owns each open check and the next status update?"

### **STATE THE NEXT UPDATE**

"We will report check status and slot confirmation separately."

### **PRESERVE THE DECISION ROUTE**

"Any exception must follow the applicable authorization process."

### **AVOID A DOWNSTREAM PROMISE**

"We cannot infer finished samples from a tape-out target."

### **RECORD THE DEPENDENCY**

"The earlier request depends on both technical readiness and capacity."

### **CLOSE CONDITIONALLY**

"October 8 remains a request pending the required evidence and acceptance."

08

## **Clean**

State the check scope, version, and remaining exceptions.

## **Waived**

Requires an actual authorized exception, not a schedule preference.

## **Committed**

Name the deliverable, date, dependencies, and accepting party.

# Choose the precise message

## **D** Check your understanding

Circle one best answer for each item. Use the stated purpose and case facts, not the option that merely sounds most polite.

### 1. Which reply to the earlier-date request is supported?

- A** October 8 remains conditional on check closure, the required handoff, and capacity confirmation.
- B** October 8 guarantees finished samples because a slot was mentioned.
- C** The unresolved findings disappear if the customer accepts schedule risk.
- D** The foundry has already reserved capacity by discussing it.

08

### 2. What does an LVS mismatch concern?

- A** A difference between extracted layout connectivity and the schematic
- B** A minimum geometric spacing violation against a design-rule deck
- C** A negative setup-slack result in a required timing scenario
- D** An offset between patterned wafer layers measured as overlay

### 3. Why retain the PDK and rule-deck versions with results?

- A** The latest available deck replaces the agreed deck without an impact review.
- B** The check result depends on the process-specific definitions actually used.
- C** A clean result from one revision automatically clears later rule changes.
- D** The design filename alone identifies every rule and model used.

### 4. Which statement about tape-out is accurate?

- A** It always occurs on the same day as customer shipment.
- B** It guarantees that no respin can be needed.
- C** It is a design-data handoff milestone, not proof of wafer start or sample availability.
- D** It means every package has completed reliability qualification.

Answers and explanations: page 99.

# A possible slot is not tape-out readiness

## E Conversations / turns 1-10

Complete the numbered gaps. Use each bank entry once, without changing its form. Cover the answers at the end until you finish.

### WORD + PHRASE BANK

delivery commitment / tape-out / sign-off / DRC / LVS / capacity reservation / waiver / PDK / Data acceptance / wafer start

08

- 01 **Sofia:** Can we move tape-out from October fifteenth to October eighth? The foundry mentioned an earlier slot, and the customer is asking whether that means samples can arrive sooner.
- 02 **Ken:** Separate the <sup>1</sup> \_\_\_\_\_ request from the later milestones. An earlier data handoff may matter to the plan, but it does not itself establish a wafer start or sample-delivery date.
- 03 **Sofia:** We still have four design-rule findings and one layout-versus-schematic mismatch. I want to know whether those can remain open while we secure the opportunity.
- 04 **Ken:** The <sup>2</sup> \_\_\_\_\_ package is not complete. No waiver is approved, so we cannot describe the design as ready by treating the open checks as an informal schedule exception.
- 05 **Sofia:** The verification team is working against P3, the agreed kit revision. We should preserve that basis when reporting progress rather than mix results from different revisions.
- 06 **Ken:** Keep the <sup>3</sup> \_\_\_\_\_ revision with the dataset and check results. The design rules and supported flow are process-specific, so version identity is part of the evidence.
- 07 **Sofia:** Please list the owner and status for each of the four findings. A falling count is not useful if the items were only relabeled.
- 08 **Ken:** Report each <sup>4</sup> \_\_\_\_\_ finding with its status under the agreed process. Do not call the run clean while unresolved findings or required decisions remain outside the summary.
- 09 **Sofia:** The connectivity mismatch is a separate issue. I do not want the customer to hear that all five items are the same kind of geometric spacing check.
- 10 **Ken:** Correct. <sup>5</sup> \_\_\_\_\_ addresses layout versus schematic consistency. Keep that distinction clear when assigning ownership and explaining why the complete sign-off package is not yet available.

# The conversation, continued

## E Conversations / turns 11-20

Continue with the word bank on page 73.

- 11 **Sofia:** On capacity, the foundry said a slot might become available. We have no written confirmation that it is allocated to this design.
- 12 **Ken:** Then there is no confirmed <sup>6</sup> \_\_\_\_\_. We can discuss feasibility, but a mentioned opportunity must not be reported as an accepted manufacturing commitment.
- 13 **Sofia:** Who confirms that the foundry has accepted the data package? I do not want a file-transfer receipt reported as acceptance of the design handoff.
- 14 **Ken:** Exactly. <sup>7</sup> \_\_\_\_\_ is a separate handoff status. Record what the foundry needs and who confirms it, rather than assume a sent dataset has cleared the receiving review.
- 15 **Sofia:** Could a formal exception be considered if one finding remains? I want to ask through the correct route, not imply that my request already authorizes it.
- 16 **Ken:** A <sup>8</sup> \_\_\_\_\_ would need the applicable technical assessment and authorization. We cannot assume it is available or accepted simply because the earlier date is commercially attractive.
- 17 **Sofia:** I will avoid calling October eighth a fabrication start. The manufacturing schedule also depends on accepted data and actual capacity arrangements beyond our requested handoff.
- 18 **Ken:** Correct. A <sup>9</sup> \_\_\_\_\_ is a different milestone. Its timing and the subsequent flow must be confirmed on their own basis before anyone promises finished samples.
- 19 **Sofia:** The next update will separate check closure from slot confirmation. It will identify owners and show what remains conditional rather than announce the earlier date as secured.
- 20 **Ken:** That protects the <sup>10</sup> \_\_\_\_\_. We can pursue the opportunity actively while making clear which evidence, acceptance, and manufacturing decisions still stand between a request and a confirmed outcome.

08

### ANSWERS / EXPLANATIONS: PAGE 100

1. tape-out | 2. sign-off | 3. PDK | 4. DRC | 5. LVS | 6. capacity reservation | 7. Data acceptance | 8. waiver | 9. wafer start | 10. delivery commitment

# A requested handoff is not a shipment

## F Say it

A team requests tape-out on November 3. Two DRC items remain open. The foundry has mentioned a possible slot but has not reserved it. No waiver or sample-delivery date is approved.

Complete the four gaps. Use each bank entry once.

### WORD + PHRASE BANK

wafer start / capacity reservation / tape-out / delivery commitment

08

Designer: "Keep November third attached to the requested design-data handoff:

<sup>1</sup> \_\_\_\_\_."

Liaison: "The possible slot is not yet an allocated <sup>2</sup> \_\_\_\_\_."

Designer: "I will not tell the customer that fabrication begins then; we have no confirmed

<sup>3</sup> \_\_\_\_\_."

Liaison: "And discussing a slot does not establish a finished-sample

<sup>4</sup> \_\_\_\_\_."

### ANSWERS / EXPLANATIONS: PAGE 100

1. tape-out | 2. capacity reservation | 3. wafer start | 4. delivery commitment

## Rehearse with a fixed script

1. Check the answers. Read the requested tape-out date separately from wafer start and finished samples.
2. Switch roles for turns 5-16. Expand PDK, DRC, and LVS using the vocabulary pages and preserve each open status.
3. Complete and check the transfer. Read the four milestones without converting a possible slot into a commitment.

### WORKING ALONE

Read both roles, pausing for two seconds when the speaker changes. Repeat with the bank covered, then check any missed expressions.

# A failing wafer map may also tell a contact story

## E More workplace conversations / 1-10

**Discuss wafer-test evidence while distinguishing a test result, a contact hypothesis, and approved retest.**

A wafer-sort run shows repeated failures at one probe site. The test program is unchanged, but contact-resistance information and probe-card history have not yet been reviewed. The test and yield engineers investigate. No cause, retest disposition, or shipment decision is established.

PEOPLE / Luca: Test engineer | Nari: Yield engineer

Use each bank entry once, without changing its form. Answers follow turn 20.

09

### WORD + PHRASE BANK

test bin / probe card / contact resistance / denominator / retest / site correlation

- 01 **Nari:** The wafer map has a repeated pattern at one probe site. Can we tell the fab that this is a process defect?
- 02 **Luca:** Not yet. First separate the <sup>1</sup> \_\_\_\_\_ from the cause. The bin tells us which test outcome was recorded, not why it happened.
- 03 **Nari:** The pattern seems to follow the same site in successive touchdowns. That could be different from a defect following a physical wafer region.
- 04 **Luca:** Exactly. Compare the <sup>2</sup> \_\_\_\_\_ with the die coordinates. Keep both views so we do not mistake the test arrangement for a wafer-location effect.
- 05 **Nari:** The program revision is unchanged. Does that rule out the test system as a contributor?
- 06 **Luca:** No. The <sup>3</sup> \_\_\_\_\_, tester configuration, connections, and conditions still need review. A matching program name does not establish identical test behavior.
- 07 **Nari:** The operations note mentions intermittent contact. I would like to see the actual measurements rather than use that phrase as the explanation.
- 08 **Luca:** I will retrieve the <sup>4</sup> \_\_\_\_\_ records and the card history. At present, contact is a hypothesis, not a confirmed cause.
- 09 **Nari:** Please preserve the first-pass results too. If the team reruns selected die later, I do not want the original failures to vanish from the dataset.
- 10 **Luca:** Agreed. Any <sup>5</sup> \_\_\_\_\_ needs the authorized plan, traceable conditions, and a separate result. Repeating until something passes is not a justified investigation.

# The conversation, continued

## E More workplace conversations / 11-20

Continue with the word bank on page 76.

- 11 **Nari:** Would a higher pass count after an intervention establish that every first-pass failure was only a contact problem?
- 12 **Luca:** No. We would need to compare the relevant evidence and understand the intervention. A changed result is useful, but it does not assign one cause to every failing die.
- 13 **Nari:** We should show first-pass yield and any later disposition separately. Otherwise the apparent improvement could simply reflect a different reporting rule.
- 14 **Luca:** Yes. Keep the <sup>6</sup> \_\_\_\_\_ and population explicit, especially if only selected die are retested.
- 15 **Nari:** I can compare the pattern across coordinates, sites, and lots. Is there anything you need from yield analysis before the equipment review?
- 16 **Luca:** The exact failing test identifiers and map references. That will let the equipment team connect its records to the same observations instead of a screenshot without context.
- 17 **Nari:** I will send those and keep the lot status unchanged. The customer update can say a site-associated pattern is under investigation.
- 18 **Luca:** Good. Do not say the fab caused it or the test system cleared it while those conclusions remain open.
- 19 **Nari:** Once the review establishes the next authorized work, we can report what was checked, what changed, and which conclusions the evidence supports.
- 20 **Luca:** Exactly. A useful wafer map leads to a focused investigation, not an immediate choice between blaming the process and discarding the test result.

09

### ANSWERS / EXPLANATIONS: PAGE 101

1. test bin | 2. site correlation | 3. probe card | 4. contact resistance | 5. retest | 6. denominator

# A selected retest is reported as the whole wafer yield

## F Say it in a new situation

Only 20 initially failing die are retested. Fifteen pass on that retest. The complete first-pass wafer results remain available.

Complete the four gaps. Use each bank entry once.

### WORD + PHRASE BANK

results / population / plan / yield

09

Yield engineer: Fifteen of twenty describes the selected retest <sup>1</sup> \_\_\_\_\_.

Test engineer: Preserve the original first-pass <sup>2</sup> \_\_\_\_\_.

Yield engineer: Do not label seventy-five percent as the whole-wafer  
<sup>3</sup> \_\_\_\_\_.

Test engineer: Interpret any changed result under the authorized retest  
<sup>4</sup> \_\_\_\_\_.

### ANSWERS / EXPLANATIONS: PAGE 101

1. population | 2. results | 3. yield | 4. plan

## Read, switch, repeat

1. Check the four answers and read the complete exchange aloud.
2. Switch roles. Read it again with the word bank covered.
3. Return to the extended conversation. Read turns 1-10, then switch roles for turns 11-20. Use the corrected expressions.

### SOLO PRACTICE

Read both roles. Pause briefly at each change of speaker. Repeat any line that needed an answer check.

# Nominal timing passes, but signoff still has an open corner

## **E** More workplace conversations / 1-10

**Explain a timing violation and challenge an unjustified exception without confusing analysis scopes.**

A digital design passes the reviewed nominal setup analysis. One required slow-corner scenario reports setup slack of -0.08 ns, and required hold analysis is incomplete. A schedule note says timing clean. The timing lead and design engineer review the claim; no exception or signoff decision has been approved.

PEOPLE / Rhea: Timing lead | Jon: Design engineer

Use each bank entry once, without changing its form. Answers follow turn 20.

10

### WORD + PHRASE BANK

Timing closure / setup slack / parasitic extraction / hold analysis / PVT corner / timing exception

- 01 **Jon:** Nominal timing passes, so I marked the design timing clean. You flagged the note. Which part of the signoff package is still open?
- 02 **Rhea:** The required slow-corner scenario has negative <sup>1</sup> \_\_\_\_\_ of zero point zero eight nanoseconds. Required hold analysis is also incomplete.
- 03 **Jon:** So the nominal result is useful, but it does not cover all the required operating assumptions. We need to show the failing scenario separately.
- 04 **Rhea:** Yes. A <sup>2</sup> \_\_\_\_\_ defines process, voltage, and temperature assumptions. Those conditions are part of the result, not a footnote we can drop.
- 05 **Jon:** The path is only used in a special mode. Could I declare it a false path to remove the violation from the report?
- 06 **Rhea:** Only with a valid, reviewed <sup>3</sup> \_\_\_\_\_. The fact that a path is inconvenient or infrequent does not prove it cannot be exercised under the relevant mode.
- 07 **Jon:** I will check the mode and constraints with verification. A false-path claim needs support from actual functional behavior, not the launch date.
- 08 **Rhea:** Exactly. Keep the endpoint, mode, and constraint reference with the question. We need to know whether the analysis is appropriate before deciding what design change is needed.
- 09 **Jon:** If we improve this setup path, can the same change affect a hold check elsewhere?
- 10 **Rhea:** It can. The <sup>4</sup> \_\_\_\_\_ must be assessed on its own required basis. We should not assume one favorable change resolves every timing concern.

# The conversation, continued

## E More workplace conversations / 11-20

Continue with the word bank on page 79.

- 11 **Jon:** The latest report uses an updated netlist, but I am not sure the extracted interconnect data belongs to that revision.
- 12 **Rhea:** Confirm the <sup>5</sup> \_\_\_\_\_ and other inputs before relying on the result. A precise slack number is not meaningful if the input configuration is inconsistent.
- 13 **Jon:** Then the next update needs the exact netlist, constraints, libraries, extraction, modes, and corners. That is more useful than another screenshot of a green summary.
- 14 **Rhea:** Yes. It also needs the open checks and reviewed exceptions.  
<sup>6</sup> \_\_\_\_\_ is against the required scope, not whichever report happens to be most favorable.
- 15 **Jon:** I will remove timing clean and report nominal setup passed, one required setup violation open, and hold checks incomplete.
- 16 **Rhea:** That preserves the completed work without implying signoff. We can then assign the constraint review and the technical investigation to the right owners.
- 17 **Jon:** Should I promise a frequency reduction as the solution? It would change the customer requirement, so I think that belongs in a separate trade-off discussion.
- 18 **Rhea:** Correct. Do not quietly change the requirement to make the report pass. Any proposed specification change needs its own assessment and authorization.
- 19 **Jon:** I will bring the reviewed path evidence to the next meeting. We will distinguish a real design issue from a justified constraint correction rather than assume either one.
- 20 **Rhea:** Good. The final decision needs consistent inputs, completed required checks, and a supported treatment of exceptions, not a nominal pass expanded into a blanket claim.

10

### ANSWERS / EXPLANATIONS: PAGE 102

1. setup slack | 2. PVT corner | 3. timing exception | 4. hold analysis | 5. parasitic extraction | 6. Timing closure

# A negative slack is hidden by an unsupported exception

## **F** Say it in a new situation

A required timing scenario has negative slack. An engineer proposes excluding the path without evidence that the exception is functionally valid.

Complete the four gaps. Use each bank entry once.

### WORD + PHRASE BANK

open / revision / mode / basis

10

Reviewer: Keep the violation visible until the exception has a justified

<sup>1</sup> \_\_\_\_\_.

Engineer: Check the path against the relevant operating <sup>2</sup> \_\_\_\_\_.

Reviewer: Preserve the actual analysis inputs and configuration

<sup>3</sup> \_\_\_\_\_.

Engineer: Do not call timing closed while required checks remain

<sup>4</sup> \_\_\_\_\_.

### ANSWERS / EXPLANATIONS: PAGE 102

1. basis | 2. mode | 3. revision | 4. open

## Read, switch, repeat

1. Check the four answers and read the complete exchange aloud.
2. Switch roles. Read it again with the word bank covered.
3. Return to the extended conversation. Read turns 1-10, then switch roles for turns 11-20. Use the corrected expressions.

### SOLO PRACTICE

Read both roles. Pause briefly at each change of speaker. Repeat any line that needed an answer check.

# A resealed bag does not explain the exposure history

## E More workplace conversations / 1-10

### Clarify moisture-handling records and distinguish shelf life, floor life, and disposition.

A reel of moisture-sensitive packaged devices arrives at assembly in a resealed bag. The original opening time and cumulative exposure record are missing. The package label identifies the part and moisture sensitivity level, but no approved disposition is recorded. A materials coordinator asks a packaging engineer how to describe the status.

PEOPLE / Sora: Materials coordinator | Caleb: Packaging engineer

Use each bank entry once, without changing its form. Answers follow turn 20.

#### WORD + PHRASE BANK

disposition / humidity indicator card / Shelf life / moisture sensitivity level / floor life / cumulative exposure

11

- 01 **Sora:** The reel is back in a sealed bag. Can I mark it ready for assembly, or do you still need the missing opening-time record?
- 02 **Caleb:** We need the history. The <sup>1</sup> \_\_\_\_\_ on the label does not tell us how this particular reel was handled after opening.
- 03 **Sora:** The bag has a recent resealing date. Someone suggested using that as a fresh start for the clock.
- 04 **Caleb:** Resealing alone does not establish a reset of <sup>2</sup> \_\_\_\_\_. We must follow the applicable product and handling requirements, not invent a new allowance.
- 05 **Sora:** I will look for the material movement records and ask the previous shift when the bag was first opened.
- 06 **Caleb:** Please distinguish a confirmed record from an estimate. We need the <sup>3</sup> \_\_\_\_\_ and storage conditions, including any interval we cannot account for.
- 07 **Sora:** There is a humidity indicator card in the bag. Does its current appearance prove that the reel was always stored correctly?
- 08 **Caleb:** No. The <sup>4</sup> \_\_\_\_\_ is one part of the applicable assessment. It is not a complete time history of every earlier handling event.
- 09 **Sora:** The supplier also gives shelf-life information. I should not use the remaining months there as an answer to the missing floor-life record.
- 10 **Caleb:** Correct. <sup>5</sup> \_\_\_\_\_ and post-opening exposure are different questions. Keep the packaging and storage conditions attached to either statement.

# The conversation, continued

## E More workplace conversations / 11-20

Continue with the word bank on page 82.

- 11 **Sora:** Could the team simply choose a baking cycle from a similar component? I do not want a language misunderstanding to become an improvised process instruction.
- 12 **Caleb:** No. Any conditioning must follow the applicable approved requirements for the actual part, package, and materials. This conversation does not prescribe a temperature or duration.
- 13 **Sora:** Then I will keep the reel identified and unavailable for routine issue while the responsible team reviews the missing history under our procedure.
- 14 **Caleb:** Yes. The <sup>6</sup> \_\_\_\_\_ must be recorded. A neat bag and a current label are not the same as an authorized material decision.
- 15 **Sora:** I can give production an update now: exact reel identified, exposure history incomplete, review requested. I should not call the devices damaged either.
- 16 **Caleb:** That is accurate. Missing evidence means the status is unresolved; it does not establish a physical failure in every device.
- 17 **Sora:** If we find the original records, I will send them with their source and timestamps rather than replacing the unknown interval with a recollection.
- 18 **Caleb:** Good. Preserve the difference between the recovered record and anyone's estimate so the reviewer can assess the actual basis.
- 19 **Sora:** I will also ask the team to improve the handover field for opening time. We should not face the same missing-history question on the next reel.
- 20 **Caleb:** Agreed. Resolve this reel through the approved process, then address the record gap without claiming that a better form retrospectively proves its handling was acceptable.

11

### ANSWERS / EXPLANATIONS: PAGE 103

1. moisture sensitivity level | 2. floor life | 3. cumulative exposure | 4. humidity indicator card | 5. Shelf life | 6. disposition

# Shelf life is used to dismiss an unknown exposure interval

## **F** Say it in a new situation

A device reel is within its stated storage shelf life, but its post-opening exposure history is incomplete.

Complete the four gaps. Use each bank entry once.

### WORD + PHRASE BANK

issue / record / history / requirements

Coordinator: Storage shelf life does not resolve the missing exposure

<sup>1</sup> \_\_\_\_\_.

Engineer: Identify the actual part, package, and applicable handling

<sup>2</sup> \_\_\_\_\_.

Coordinator: Keep unknown time separate from a confirmed <sup>3</sup> \_\_\_\_\_.

Engineer: Await the authorized material decision before routine

<sup>4</sup> \_\_\_\_\_.

11

### ANSWERS / EXPLANATIONS: PAGE 103

1. history | 2. requirements | 3. record | 4. issue

## Read, switch, repeat

1. Check the four answers and read the complete exchange aloud.
2. Switch roles. Read it again with the word bank covered.
3. Return to the extended conversation. Read turns 1-10, then switch roles for turns 11-20. Use the corrected expressions.

### SOLO PRACTICE

Read both roles. Pause briefly at each change of speaker. Repeat any line that needed an answer check.

# 01 Briefing and language checks

## Wafer Fabrication Flow and Process Integration

### A Briefing checks / page 4

1. Where is L24 now?

**D. In wafer fabrication at an M2-related operation**

The briefing places L24 in wafer fabrication, not the packaging stage Eric inferred.

2. What does BEOL mean in this case?

**A. Back end of line, forming on-wafer interconnects**

The integration engineer uses BEOL for on-wafer interconnect fabrication.

3. What prevents the next operation?

**C. A pending inspection review and lot hold**

The lot remains on hold until the stated inspection review and authorized disposition.

### D Language checks / page 9

1. Which customer update is accurate?

**A. L24 remains in wafer fabrication at M2, on hold for inspection review; packaging has not started.**

M2 identifies the current fabrication work, not completion of the stack or assembly. The hold still prevents an unapproved next move.

2. Why expand BEOL for the coordinator?

**D. Different teams can use back-end for different parts of the manufacturing chain.**

Expanding the term prevents a cross-team ambiguity that already caused an inaccurate update.

3. What is the appropriate next-step wording?

**B. The lot awaits inspection review and authorized disposition under its route.**

The stated hold and route control what may happen next, not a generic assumption about progress.

4. Which distinction is correct?

**A. Wafer sort tests die on the wafer; package assembly integrates die into packages.**

Wafer sort is electrical testing, dicing is physical separation, FEOL principally forms devices, and BEOL forms on-wafer interconnects.

# 01 Conversation answers

## Wafer Fabrication Flow and Process Integration

### E Dialogue gaps

Pages 10-11

- 1. BEOL.** Turn 2. BEOL refers to the on-wafer interconnect stage in this case, not to completed package assembly.
- 2. wafer fabrication.** Turn 4. Wafer fabrication locates the current work on the wafer before the package assembly stage described in the mistaken update.
- 3. metal layer.** Turn 6. The metal layer identifies the specific interconnect level involved in the current operation.
- 4. FEOL.** Turn 8. FEOL principally forms active devices, distinct from the later on-wafer interconnect work now involving L24.
- 5. lot hold.** Turn 10. A lot hold restricts further movement until the required review and authorized disposition occur.
- 6. disposition.** Turn 12. Disposition is the authorized decision about how the lot may proceed after review, not an assumption based on its position in the flow.
- 7. traveler.** Turn 14. The traveler connects the material with its operation history and required next steps, supporting traceable status communication.
- 8. Wafer sort.** Turn 16. Wafer sort is electrical testing of die on the wafer and is distinct from package assembly or shipment.
- 9. package assembly.** Turn 18. Package assembly is the later integration of die into packages, which has not started for L24.
- 10. process flow.** Turn 20. The process flow connects the current stage and authorized next steps without collapsing fabrication, testing, and assembly into one label.

### F Transfer conversation

Page 12

#### 1. BEOL

Coordinator: "I will correct the update: this on-wafer interconnect work is still BEOL."

BEOL names the on-wafer interconnect stage, not package assembly.

#### 2. lot hold

Engineer: "Keep the restriction visible too; the record still shows a lot hold."

A lot hold restricts movement while the required review and decision remain open.

#### 3. package assembly

Coordinator: "Then I should not tell the customer we have started package assembly."

Package assembly is distinct from the current on-wafer work and has not begun here.

#### 4. disposition

Engineer: "Correct. Ask the review owner for the authorized disposition before discussing the next move."

Disposition is the authorized material decision, not a date inferred from the process stage.

# 02 Briefing and language checks

## Lithography, Reticles, and Critical Dimensions

### A Briefing checks / page 13

1. Which layer and stage do the records identify?

**D. M2 after etch**

The recovered records specifically locate the two means on M2 after etch.

2. What is the reported numerical difference?

**C. An increase of 3 nm**

Subtracting the first mean, 86 nm, from the second, 89 nm, gives a 3 nm increase.

3. Why is a direct process attribution premature?

**C. The measurement recipes and sampling sites differ.**

Changed measurement and sampling conditions prevent the difference from isolating a process cause.

### D Language checks / page 18

1. Which report sentence is justified?

**C. The means differ by 3 nm, but recipe and site changes limit comparability.**

The sentence reports the arithmetic while preserving the unresolved measurement basis.

2. Which comparison best helps investigate the apparent change?

**C. Comparable measurements on identified sites using an agreed recipe basis**

An agreed measurement basis addresses the changed recipes and sampling. More data alone does not remove those differences, and different stages answer different questions.

3. Why is post-etch important?

**B. The result follows pattern transfer and does not isolate lithography alone.**

The measurement stage affects interpretation because more than the exposure step contributes to the resulting feature.

4. Which distinction is accurate?

**A. CD concerns feature size; overlay concerns layer alignment.**

CD and overlay separate size from alignment. Pitch is repeat distance, roughness is edge variation, dose is exposure energy per area, and etch bias concerns dimensional transfer.

# 02 Conversation answers

## Lithography, Reticles, and Critical Dimensions

### E Dialogue gaps

Pages 19-20

- 1. post-etch.** Turn 2. Post-etch locates the measurement after pattern transfer, which matters when interpreting possible causes.
- 2. measurement recipe.** Turn 4. The measurement recipe defines settings and analysis that can affect the reported CD and differs between these datasets.
- 3. sampling site.** Turn 6. A sampling site is a selected measurement location; different site selections can affect the reported average.
- 4. critical dimension.** Turn 8. Critical dimension is the relevant feature size, but its reported change does not alone establish the physical cause.
- 5. Etch bias.** Turn 10. Etch bias concerns the dimensional change associated with pattern transfer and is one reason post-etch CD does not isolate exposure.
- 6. Overlay.** Turn 12. Overlay describes alignment between layers, whereas CD describes a feature dimension.
- 7. remeasurement.** Turn 14. Remeasurement on an agreed basis can help determine whether the apparent difference persists under comparable conditions.
- 8. measurement bias.** Turn 16. Measurement bias is a systematic difference in measurement relative to a reference and is distinct from an assumed process shift.
- 9. reticle.** Turn 18. The reticle carries the lithographic pattern; identifying its revision provides context without proving it caused the observation.
- 10. pattern fidelity.** Turn 20. Pattern fidelity concerns how well the produced pattern matches its intent; assessing it requires more than an unexplained mean difference.

### F Transfer conversation

Page 21

#### 1. critical dimension

Engineer: "Check that both reports refer to the same critical dimension."

Critical dimension refers to feature size, not the settings or locations used to measure it.

#### 2. measurement recipe

Reviewer: "Send the settings and analysis from each measurement recipe."

The recipe defines the settings and analysis that differ between the datasets.

#### 3. sampling sites

Engineer: "I will attach a wafer map of both sets of sampling sites."

Sampling sites identify the measurement locations, which also differed in this case.

#### 4. overlay

Reviewer: "Good. An alignment claim would need separate overlay evidence."

Overlay addresses alignment rather than the reported feature-size difference of four nanometers.

## 03 Briefing and language checks

### Deposition, Etch, CMP, and Process Windows

03

#### A Briefing checks / page 22

1. What improved in the supplied trial?

**B. The stated nonuniformity metric fell from 5% to 3% at the nominal setting.**

The only reported improvement concerns the unchanged metric at one tested condition.

2. What is the absolute change in that percentage metric?

**A. A decrease of 2 percentage points**

Five percent minus three percent equals two percentage points on the stated metric.

3. Which conclusion is premature?

**C. The complete process window is validated.**

The broader range and downstream behavior were not assessed, so full-window validation is unsupported.

#### D Language checks / page 27

1. Which summary preserves the evidence boundary?

**B. The nominal-setting metric improved; the range and downstream effects remain untested.**

The observed improvement concerns one response at one setting. Untested settings, other film properties, integration, and release remain separate questions.

2. What is the relative reduction from 5% to 3%?

**C. 40%**

The decrease of two divided by the original five equals forty percent.

3. Why review downstream etch and CMP?

**B. A film change may alter later process behavior.**

A changed film can interact differently with later processes. A shared recipe or one improved response does not establish selectivity, polishing behavior, or all material properties.

4. Which status should remain separate from trial success?

**B. Recipe release**

Recipe release is an authorized decision that has not occurred in this case.

# 03 Conversation answers

## Deposition, Etch, CMP, and Process Windows

03

### E Dialogue gaps

Pages 28-29

- 1. process window.** Turn 2. A process window is a range satisfying defined requirements, not a single successful nominal setting.
- 2. thickness nonuniformity.** Turn 4. Thickness nonuniformity is the specific response that improved; it does not stand for all film or integration properties.
- 3. nominal setting.** Turn 6. The nominal setting is the only process condition tested, so the measured improvement must remain limited to it.
- 4. corner condition.** Turn 8. A corner condition is a defined limiting setting or combination that requires its own evidence in the evaluation plan.
- 5. integration trade-off.** Turn 10. An integration trade-off concerns benefits and consequences across related steps that a single improved deposition metric cannot settle.
- 6. etch selectivity.** Turn 12. Etch selectivity compares material removal rates and may be affected by a film change; no result is supplied here.
- 7. CMP.** Turn 14. CMP is chemical mechanical planarization, a downstream process whose compatibility with the changed film remains unassessed.
- 8. process interaction.** Turn 16. A process interaction is a relationship between factors or steps that can change the overall result and needs explicit assessment.
- 9. recipe release.** Turn 18. Recipe release is the controlled authorization for defined use, separate from a promising experimental result.
- 10. deposition.** Turn 20. Deposition is the film-forming step actually assessed; the statement should not imply the entire integrated flow was validated.

### F Transfer conversation

Page 30

#### 1. nominal setting

Engineer: "The metric fell from eight to six percent, but only at the nominal setting."

The nominal setting is the single tested condition, not the full range.

#### 2. process window

Reviewer: "Keep the low and high settings open; we have not established the process window."

A process window requires evidence across the relevant range, which is missing here.

#### 3. integration question

Engineer: "I will also list the untested effect on the later etch as an integration question."

The film change may affect a later step, so downstream compatibility remains an integration question.

#### 4. recipe release

Reviewer: "And keep routine production use pending the authorized recipe release."

A favorable trial result does not itself authorize the recipe for production use.

# 04 Briefing and language checks

## Metrology, SPC, and Yield Learning

04

### A Briefing checks / page 31

1. What is the pilot yield?

**B. 92% from 92 of 100 tested die**

Ninety-two passing die divided by one hundred tested gives a 92% observed yield.

2. What is the absolute difference between the observed yields?

**C. 2 percentage points**

The observed yields are 92% and 90%, a difference of two percentage points.

3. Which factor limits the causal comparison?

**C. The product mixes differ.**

Different product mixes can affect yield, so the process change is not isolated by this comparison.

### D Language checks / page 36

1. Which announcement is supported?

**D. The pilot observed 92% versus 90% yield, with different product mixes and no causal conclusion.**

The counts support descriptive percentages, but the comparison does not isolate cause or establish stability.

2. What is the approximate relative increase from 90% to 92%?

**B. 2.22%**

The two-point increase divided by the 90% starting value is approximately 2.22%.

3. Which next analysis addresses the product-mix problem?

**B. Separate comparable product groups and review consistent test bases.**

Stratification examines comparable product groups. Equal sample sizes, pooled totals, or pass counts alone do not remove a difference in product composition.

4. Which statement distinguishes control and specification limits?

**A. Control limits monitor behavior; specification limits define acceptance boundaries.**

Statistical stability and conformity are different questions. A stable process may miss specifications, and in-spec observations do not establish statistical control.

# 04 Conversation answers

## Metrology, SPC, and Yield Learning

### E Dialogue gaps

Pages 37-38

**1. pass count.** Turn 2. The pass count identifies the actual successes behind the percentage, preserving the scale of the evidence.

**2. product mix.** Turn 4. Product mix differs between the lots and may affect observed yield independently of the process change.

**3. percentage point.** Turn 6. Percentage point measures the absolute difference between the two yield percentages, here 92 minus 90.

**4. causal attribution.** Turn 8. Causal attribution assigns the observed yield difference to the process change, which this mixed comparison does not isolate.

**5. stratification.** Turn 10. Stratification separates the data into comparable product groups to examine the mix effect more directly.

**6. process stability.** Turn 12. Process stability concerns statistical behavior over time under a defined monitoring method, not two favorable aggregate yields.

**7. specification limit.** Turn 14. A specification limit defines acceptance requirements and is distinct from a statistically based process-monitoring boundary.

**8. control signal.** Turn 16. A control signal triggers investigation under the monitoring method; it does not provide its own causal diagnosis.

**9. measurement system.** Turn 18. The measurement system includes methods and conditions that affect whether the collected results can be compared consistently.

**10. yield learning.** Turn 20. Yield learning uses the observations to investigate losses and drivers without claiming an effect before the evidence supports it.

### F Transfer conversation

Page 39

#### 1. denominator

Engineer: "Put 48 passing die over the trial's 50-die denominator."

The tested total is the denominator; the 48 passing die form the numerator.

#### 2. product mix

Reviewer: "Then show the changed product mix beside the reference comparison, not in a separate appendix."

Product mix differs between the trial and reference and may affect the observed yields.

#### 3. percentage points

Engineer: "I will report the difference as six percentage points, not six relative percent."

Ninety-six percent minus ninety percent is six percentage points, not six relative percent.

#### 4. causal attribution

Reviewer: "Good. Leave the claim that the process caused it open until we have supported causal attribution."

Causal attribution needs evidence isolating the process effect, which the supplied comparison lacks.

# 05 Briefing and language checks

## Defect Density and Cleanroom Contamination

### A Briefing checks / page 40

1. What does the inspection record contain?

**A. 24 particle-like detections over 200 square centimeters**

The record describes surface inspection detections and inspected area, not confirmed failure or added-defect counts.

2. What does the map show?

**D. An edge cluster**

The supplied observation is a spatial edge cluster, without an established source.

3. Why cannot the 24 detections be called 24 added defects?

**C. No comparable pre-process scan exists, and review is incomplete.**

Added defects require a valid before-and-after basis, and the detections are not all confirmed defects.

### D Language checks / page 45

1. What is the observed detection density?

**B. 0.12 detections per square centimeter**

Twenty-four detections divided by two hundred square centimeters equals 0.12 on a surface-area basis.

2. Which conclusion is unsupported?

**D. C2 introduced all 24 detections.**

Timing after C2 does not establish introduction by C2 without the needed comparison and investigation.

3. Which next record is useful for the investigation?

**C. Traceable wafer, inspection-setting, timing, and exposure information**

The traceable records preserve the actual material and measurement context. Unmatched layers, aggregate totals, and unsupported conversions cannot replace that evidence.

4. Which distinction is correct?

**A. Surface detection density and airborne particle counts use different measurement bases.**

Surface and air results have different bases. Detection is not confirmed classification, timing is not proof of introduction, and a pattern alone does not identify a source.

# 05 Conversation answers

## Defect Density and Cleanroom Contamination

### E Dialogue gaps

Pages 46-47

**1. inspected area.** Turn 2. The inspected area provides the denominator for the surface detection density rather than leaving the count without scale.

**2. Defect review.** Turn 4. Defect review determines how flagged features should be classified before they are treated as confirmed relevant defects.

**3. edge cluster.** Turn 6. An edge cluster describes the spatial concentration actually observed without claiming its origin has been established.

**4. added defect.** Turn 8. An added defect requires evidence that it appeared between comparable inspection points, which is missing here.

**5. pre-process scan.** Turn 10. A pre-process scan provides the earlier observation needed for a valid before-and-after comparison when methods are comparable.

**6. airborne particle count.** Turn 12. An airborne particle count concerns a defined air sample or volume, unlike the surface-area-based detections here.

**7. inspection sensitivity.** Turn 14. Inspection sensitivity affects which features are detected and must be considered when comparing inspection results.

**8. containment scope.** Turn 16. Containment scope identifies which material or operations are restricted while the investigation remains open.

**9. exposure history.** Turn 18. Exposure history connects the wafers to relevant handling and environments that may help evaluate possible contamination origins.

**10. contamination source.** Turn 20. The contamination source is the origin to be established by investigation, not assumed from the latest process step or vague label.

### F Transfer conversation

Page 48

#### 1. inspected area

Engineer: "Keep 100 square centimeters beside the count as the inspected area."

Inspected area supplies the surface basis for the 12 detections, giving a density of 0.12.

#### 2. defect review

Reviewer: "And leave the twelve flagged features unclassified pending defect review."

Defect review determines how inspection detections should be classified; it remains incomplete.

#### 3. pre-process scan

Engineer: "I will request a comparable pre-process scan before reporting any process-added count."

A comparable pre-process scan is needed to assess what was introduced between inspection points.

#### 4. contamination source

Reviewer: "Thank you. Do not name the last tool as the contamination source without supporting investigation."

The source is an origin to investigate, not a conclusion established by the post-process count.

05

# 06 Briefing and language checks

## Equipment Uptime, Recipes, and Tool Matching

### A Briefing checks / page 49

1. How far apart are the sample means?

**D. 4 nm**

The difference between 102 nm and 98 nm is four nanometers.

2. Do the means meet the stated matching criterion?

**C. No; 4 nm exceeds the allowed 2 nm difference.**

The matching criterion applies to the difference between means, not merely each mean's specification status.

3. What does 90% describe here?

**A. Availability for 900 of the defined 1,000 minutes**

The supplied time ratio concerns equipment availability and establishes no product-yield or approval claim.

### D Language checks / page 54

1. Which status report is accurate?

**C. Both means meet the mean specification; their 4 nm difference fails the 2 nm matching limit.**

The specification and matching tests use different criteria, and individual results are not supplied.

2. What does using the same recipe establish?

**B. The identified recipe input is shared, not that output performance is interchangeable.**

Shared commanded inputs do not establish identical physical conditions, equivalent output, or authorization to transfer the process.

3. Which calculation matches the availability basis?

**C. 900 divided by 1,000 equals 90%.**

Availability is available time over the defined window. Ten percent is the unavailable fraction, and utilization needs its own use-time data and basis.

4. What remains necessary before calling B an approved interchangeable backup?

**B. The required matching or qualification evidence and transfer approval**

The case lacks matching acceptance and transfer approval, neither of which is replaced by uptime.

# 06 Conversation answers

## Equipment Uptime, Recipes, and Tool Matching

### E Dialogue gaps

Pages 55-56

**1. Tool matching.** Turn 2. Tool matching requires measured performance against a defined comparison criterion, not simply a shared recipe or uptime figure.

**2. matching criterion.** Turn 4. The matching criterion allows no more than two nanometers between means; the observed difference is four.

**3. actual value.** Turn 6. Actual value concerns the measured outcome; passing the mean specification does not remove the separate difference between tools.

**4. recipe revision.** Turn 8. The recipe revision identifies the shared controlled input, but cannot stand in for unsupplied individual wafer evidence.

**5. chamber condition.** Turn 10. Chamber condition is a possible contributor to tool behavior that must be assessed rather than assumed from the recipe.

**6. calibration status.** Turn 12. Calibration status concerns the documented measurement basis and is relevant to interpreting the observed tool difference.

**7. availability.** Turn 14. Availability is the available-time fraction on the stated basis, not a measure of matching, yield, or approval.

**8. equipment state.** Turn 16. Equipment state records the tool's operational condition; an available state does not establish qualified process performance.

**9. process transfer.** Turn 18. Process transfer moves a defined process to another tool and requires its own relevant assessment and approval.

**10. release status.** Turn 20. Release status records authorization for the intended use, which remains absent despite the tool's availability and passing means.

### F Transfer conversation

Page 57

#### 1. specification

Engineer: "Both means meet the 48-to-55 nm specification."

The specification applies to each sample mean, and both 50 and 53 fall within it.

#### 2. matching criterion

Reviewer: "But their three-nanometer gap misses our one-nanometer matching criterion."

The matching criterion applies to the three-nanometer difference, which exceeds the permitted one.

#### 3. process transfer

Engineer: "Then moving this process to the other tool remains a proposed process transfer."

Process transfer is the move between tools and requires its own assessment and authorization.

#### 4. release status

Reviewer: "Correct. Keep approval pending in the release status field."

Release status concerns authorization; passing individual means does not establish that approval.

# 07 Briefing and language checks

## Packaging, Test, and Reliability Qualification

### A Briefing checks / page 58

1. Which test has passed?

**A. Temperature cycling**

The briefing identifies temperature cycling as the only completed passing test.

2. What is the operating-life test status?

**A. Running**

The supplied plan still includes the test, and its stated status is running.

3. Why is qualification complete unsupported?

**D. Two required tests and final review remain open.**

The defined plan requires evidence and final review that have not all been completed.

### D Language checks / page 63

1. Which customer headline is supported?

**D. Qualification in progress: temperature cycling passed; HTOL running; HAST not started.**

The headline accurately separates each test's status without inventing a result or approval.

2. What does the passing temperature-cycle result establish?

**D. The stated test result under its protocol and scope**

Temperature cycling, operating-life testing, and humidity-related stress address different test conditions. A shared name also does not establish equivalent package construction or scope.

3. When could related evidence replace a required test in this case?

**D. Only through the applicable accepted change or similarity process, which has not occurred**

Related evidence requires an accepted relevance and authorization basis. Dimensions, document age, and a different passing test do not by themselves justify substitution.

4. What should final review examine?

**A. Required evidence, scope, configuration, open items, and approval conditions**

Qualification approval depends on the required evidence and scope, not a favorable summary label.

# 07 Conversation answers

## Packaging, Test, and Reliability Qualification

### E Dialogue gaps

Pages 64-65

**1. qualification plan.** Turn 2. The qualification plan defines all required tests and final review, so one passing test cannot complete it.

**2. thermal cycling.** Turn 4. Thermal cycling is the completed temperature-change stress test; its passing result remains valid within its defined protocol.

**3. HTOL.** Turn 6. HTOL is high-temperature operating life testing, which is currently running rather than completed in this case.

**4. HAST.** Turn 8. HAST is the highly accelerated stress test that has not started; absence of a result is not a pass or failure.

**5. package construction.** Turn 10. Package construction identifies the materials and structure covered by the evidence, preventing unsupported extension to other variants.

**6. qualification vehicle.** Turn 12. The qualification vehicle is the selected device or structure whose results support the defined qualification scope.

**7. preconditioning.** Turn 14. Preconditioning records show the defined preparation used before testing and form part of interpreting the evidence.

**8. Qualification by similarity.** Turn 16. Qualification by similarity requires an accepted technical rationale; no such substitution is established in this case.

**9. Test coverage.** Turn 18. Test coverage identifies which requirements or concerns the available tests address; final electrical testing does not replace every reliability stress.

**10. qualification approval.** Turn 20. Qualification approval is the authorized acceptance of the required evidence, which remains pending while tests and review are incomplete.

### F Transfer conversation

Page 66

#### 1. evidence

Engineer: "Keep A's passing result as test evidence, not a claim that every requirement is complete."

The passing result is evidence for its defined test, not completion of the whole plan.

#### 2. plan

Reviewer: "B and C are still required by the approved plan."

The approved qualification plan still requires these tests; neither has an accepted replacement.

#### 3. rationale

Engineer: "We would need an accepted technical rationale before substituting related results."

A justified and accepted rationale is necessary for a relevant substitution; none exists here.

#### 4. approval

Reviewer: "Until the tests and final review are complete, do not announce qualification approval."

Qualification approval remains pending while required tests and final review are incomplete.

# 08 Briefing and language checks

## Foundry, Tape-Out, PDK, and Capacity Communication

### A Briefing checks / page 67

1. Which earlier date is requested?

**B. October 8 tape-out**

Sofia asks to move tape-out from October 15 to October 8, not to deliver finished devices then.

2. What remains unresolved in the design checks?

**B. Four DRC findings and one LVS mismatch**

The briefing explicitly lists four design-rule findings and one layout-versus-schematic mismatch.

3. What is the capacity status?

**D. A possible earlier slot, not reserved or accepted**

A mentioned opportunity is not a recorded capacity commitment or approval.

### D Language checks / page 72

1. Which reply to the earlier-date request is supported?

**A. October 8 remains conditional on check closure, the required handoff, and capacity confirmation.**

The reply preserves the separate technical and capacity prerequisites without inventing acceptance.

2. What does an LVS mismatch concern?

**A. A difference between extracted layout connectivity and the schematic**

LVS compares the layout-derived circuit with the schematic. Spacing rules, timing slack, and overlay belong to different verification or measurement questions.

3. Why retain the PDK and rule-deck versions with results?

**B. The check result depends on the process-specific definitions actually used.**

Traceable versions identify the actual basis of the result. A filename is insufficient, and neither a later release nor an earlier clean run establishes unchanged applicability.

4. Which statement about tape-out is accurate?

**C. It is a design-data handoff milestone, not proof of wafer start or sample availability.**

Fabrication, test, qualification, and delivery remain distinct activities after the design-data handoff.

# 08 Conversation answers

## Foundry, Tape-Out, PDK, and Capacity Communication

### E Dialogue gaps

Pages 73-74

**1. tape-out.** Turn 2. Tape-out is the design-data release milestone being requested, not a confirmed wafer start or finished-sample date.

**2. sign-off.** Turn 4. Sign-off is the required completion and approval of the defined checks; the complete package is not yet available here.

**3. PDK.** Turn 6. The PDK supplies process-specific design resources; its agreed revision must remain connected to the verification results.

**4. DRC.** Turn 8. DRC checks the layout against applicable design rules; its open findings need documented resolution rather than a changed headline.

**5. LVS.** Turn 10. LVS compares extracted layout connectivity with the schematic and therefore addresses a different verification question from geometric rule checks.

**6. capacity reservation.** Turn 12. Capacity reservation is an actual allocation; a possible earlier slot has not been reserved or accepted in this case.

**7. Data acceptance.** Turn 14. Data acceptance confirms the receiving process's requirements have been met; transmitting files alone does not establish it.

**8. waiver.** Turn 16. A waiver is an authorized exception to a specific requirement; none is approved for the open findings here.

**9. wafer start.** Turn 18. A wafer start is the lot's entry into fabrication and is distinct from the earlier design-data handoff milestone.

**10. delivery commitment.** Turn 20. A delivery commitment must identify an accepted deliverable and date; neither a requested tape-out nor a possible slot guarantees finished samples.

### F Transfer conversation

Page 75

#### 1. tape-out

Designer: "Keep November third attached to the requested design-data handoff: tape-out."

Tape-out identifies the requested design-data handoff, not a later manufacturing or delivery event.

#### 2. capacity reservation

Liaison: "The possible slot is not yet an allocated capacity reservation."

Capacity reservation requires a confirmed allocation; a possible slot has not reached that status.

#### 3. wafer start

Designer: "I will not tell the customer that fabrication begins then; we have no confirmed wafer start."

A wafer start is distinct from the requested design-data handoff and is not confirmed here.

#### 4. delivery commitment

Liaison: "And discussing a slot does not establish a finished-sample delivery commitment."

A delivery commitment needs an accepted deliverable and date, neither of which follows from discussing a slot.

# 09 Conversation answers

## A failing wafer map may also tell a contact story

### E Extended conversation

Pages 76-77

**1. test bin.** Turn 2. A test bin classifies the recorded test outcome; it does not by itself identify the physical cause of a failure.

**2. site correlation.** Turn 4. Site correlation examines whether the outcome follows a particular parallel test site, helping distinguish that pattern from the wafer's physical geography.

**3. probe card.** Turn 6. The probe card provides the electrical interface to wafer contacts; its condition and history can matter even when the test-program revision is unchanged.

**4. contact resistance.** Turn 8. Contact resistance concerns the electrical resistance at the probing connection; relevant measurements can help investigate a contact hypothesis without proving it from a label.

**5. retest.** Turn 10. Retest repeats testing under the applicable authorized conditions; it must preserve the first-pass record and identify the basis for interpreting a changed result.

**6. denominator.** Turn 14. The denominator identifies the population used in a yield calculation; first-pass and selected-retest populations must not be silently interchanged.

### F Transfer exchange

Page 78

#### 1. population

Yield engineer: Fifteen of twenty describes the selected retest population.

The retest population contains only selected initially failing die, not the whole wafer.

#### 2. results

Test engineer: Preserve the original first-pass results.

Later testing must not erase the original observations or their reporting basis.

#### 3. yield

Yield engineer: Do not label seventy-five percent as the whole-wafer yield.

Fifteen divided by twenty is the selected retest pass rate, not overall wafer yield.

#### 4. plan

Test engineer: Interpret any changed result under the authorized retest plan.

The authorized plan determines the conditions and basis for using the retest evidence.

Background reading: [FormFactor: Wafer Probing and Test Community](#). The conversation is an original fictional teaching case.

# 10 Conversation answers

## Nominal timing passes, but signoff still has an open corner

### E Extended conversation

Pages 79-80

**1. setup slack.** Turn 2. Setup slack compares the required and actual arrival timing for the setup check; negative slack means the stated constraint is not met in that scenario.

**2. PVT corner.** Turn 4. A PVT corner specifies process, voltage, and temperature assumptions for analysis; a pass at one corner does not automatically establish a pass at others.

**3. timing exception.** Turn 6. A timing exception changes how specified paths are analyzed and requires a justified functional basis; it is not a way to hide an inconvenient violation.

**4. hold analysis.** Turn 10. Hold analysis checks minimum-delay timing requirements and is distinct from setup analysis; a setup improvement does not automatically establish hold correctness.

**5. parasitic extraction.** Turn 12. Parasitic extraction provides modeled interconnect effects such as resistance and capacitance; its configuration must match the design being analyzed.

**6. Timing closure.** Turn 14. Timing closure is the process of meeting the defined timing requirements across the required analysis scope, not merely obtaining one favorable nominal report.

### F Transfer exchange

Page 81

#### 1. basis

Reviewer: Keep the violation visible until the exception has a justified basis.

An exception needs a valid functional rationale rather than merely a desire to remove the violation.

#### 2. mode

Engineer: Check the path against the relevant operating mode.

Functional behavior in the applicable mode matters when deciding whether an exception is appropriate.

#### 3. revision

Reviewer: Preserve the actual analysis inputs and configuration revision.

The timing result must remain linked to the precise configuration analyzed.

#### 4. open

Engineer: Do not call timing closed while required checks remain open.

Incomplete required checks prevent a blanket claim that the defined timing scope is complete.

10

Background reading: [Synopsys: Static Timing Analysis](#). The conversation is an original fictional teaching case.

# 11 Conversation answers

## A resealed bag does not explain the exposure history

### E Extended conversation

Pages 82-83

**1. moisture sensitivity level.** Turn 2. The moisture sensitivity level identifies a package's classification for moisture-related handling; it is not evidence that this reel's actual exposure stayed within its requirements.

**2. floor life.** Turn 4. Floor life concerns the permitted exposure under specified conditions before the relevant assembly process; a new bag seal alone does not establish a fresh exposure allowance.

**3. cumulative exposure.** Turn 6. Cumulative exposure concerns the relevant accumulated time and conditions after opening; an undocumented interval cannot be treated as zero simply because the reel was resealed.

**4. humidity indicator card.** Turn 8. A humidity indicator card provides an indication relevant to humidity within the package under its specified interpretation; it does not reconstruct the complete historical exposure record.

**5. Shelf life.** Turn 10. Shelf life concerns storage over a defined period under stated conditions; it is a different question from post-opening floor life before assembly.

**6. disposition.** Turn 14. Disposition is the authorized decision about the material's next route after reviewing the relevant evidence and requirements; resealing does not supply that decision.

### F Transfer exchange

Page 84

#### 1. history

Coordinator: Storage shelf life does not resolve the missing exposure history.

The storage period and post-opening handling record address different requirements.

#### 2. requirements

Engineer: Identify the actual part, package, and applicable handling requirements.

The assessment must use the requirements for the specific material rather than a similar component.

#### 3. record

Coordinator: Keep unknown time separate from a confirmed record.

An undocumented interval must not be treated as a verified value.

#### 4. issue

Engineer: Await the authorized material decision before routine issue.

The unresolved handling status needs the applicable disposition before normal production issue.

11

Background reading: [Texas Instruments: Product Shelf Life](#). The conversation is an original fictional teaching case.

# Find the move you need

Use these as precise language models. Replace details only when you know the actual facts, authority, and agreed conditions.

## 01 Wafer Fabrication Flow and Process Integration

**Clarify the abbreviation:** Here BEOL means back end of line.

**Separate the stages:** On-wafer interconnect formation is not package assembly.

**State the hold:** L24 is on hold pending inspection review.

Full phrase set: pages 7-8 | Dialogue: pages 10-11

## 02 Lithography, Reticles, and Critical Dimensions

**Locate the dimension:** Which layer and feature does the CD refer to?

**Qualify the difference:** The apparent difference is 3 nm.

**Separate metrics:** Overlay is not another name for critical dimension.

Full phrase set: pages 16-17 | Dialogue: pages 19-20

## 03 Deposition, Etch, CMP, and Process Windows

**Identify the metric:** The defined thickness-nonuniformity metric fell from 5% to 3%.

**Separate changes:** That is two percentage points, or a 40% relative reduction.

**Ask about planarization:** Has compatibility with the CMP step been assessed?

Full phrase set: pages 25-26 | Dialogue: pages 28-29

## 04 Metrology, SPC, and Yield Learning

**Give the pilot counts:** The pilot passed 92 of 100 tested die.

**Name the relative basis:** Relative to 90%, that is approximately 2.22%.

**Request like-for-like data:** Compare relevant product groups on a consistent basis.

Full phrase set: pages 34-35 | Dialogue: pages 37-38

# Keep the conversation productive

Use these as precise language models. Replace details only when you know the actual facts, authority, and agreed conditions.

## 05 Defect Density and Cleanroom Contamination

**Replace the vague label:** The record shows particle-like surface detections, not a confirmed cleanroom source.

**Locate the pattern:** The wafer map shows an edge cluster.

**Separate measurement types:** Surface detections are not airborne particle counts.

Full phrase set: pages 43-44 | Dialogue: pages 46-47

## 06 Equipment Uptime, Recipes, and Tool Matching

**Confirm the recipe:** Both tools used the identified R7 revision.

**Give the difference:** The means are 4 nm apart.

**Ask about chamber state:** Were the relevant chamber conditions documented?

Full phrase set: pages 52-53 | Dialogue: pages 55-56

## 07 Packaging, Test, and Reliability Qualification

**Name the completed result:** Q7 passed temperature cycling under the stated protocol.

**Keep the plan visible:** The approved plan requires all three tests and final review.

**Ask about the vehicle:** Is this the approved qualification vehicle?

Full phrase set: pages 61-62 | Dialogue: pages 64-65

## 08 Foundry, Tape-Out, PDK, and Capacity Communication

**Name the requested milestone:** You are requesting October 8 tape-out.

**Confirm the basis:** The checks use the agreed PDK revision P3.

**Keep sign-off visible:** The complete sign-off package is not yet available.

Full phrase set: pages 70-71 | Dialogue: pages 73-74

# Find a term and its context

The page number leads to the definition and a typical word combination. Repeated terms have separate entries when they serve different lesson contexts.

|                                                                   |    |
|-------------------------------------------------------------------|----|
| <b>actual value</b> compare setpoint and actual value             | 50 |
| <b>added defect</b> quantify added defects                        | 41 |
| <b>airborne particle count</b> review airborne particle counts    | 42 |
| <b>ALD</b> assess ALD conformance                                 | 23 |
| <b>alignment mark</b> inspect alignment marks                     | 15 |
| <b>anisotropy</b> evaluate etch anisotropy                        | 23 |
| <b>availability</b> calculate equipment availability              | 50 |
| <b>backup tool</b> qualify a backup tool                          | 51 |
| <b>BEOL</b> track BEOL progress                                   | 5  |
| <b>BGA</b> assess BGA assembly                                    | 59 |
| <b>calibration status</b> check calibration status                | 51 |
| <b>capacity reservation</b> confirm capacity reservation          | 69 |
| <b>causal attribution</b> qualify causal attribution              | 33 |
| <b>CD-SEM</b> verify the CD-SEM recipe                            | 14 |
| <b>chamber condition</b> assess chamber condition                 | 51 |
| <b>chamber matching</b> verify chamber matching                   | 50 |
| <b>CMP</b> evaluate CMP compatibility                             | 24 |
| <b>common-cause variation</b> characterize common-cause variation | 33 |
| <b>conformality</b> assess film conformality                      | 23 |
| <b>confounding factor</b> identify a confounding factor           | 32 |
| <b>containment scope</b> define containment scope                 | 42 |
| <b>contamination source</b> identify the contamination source     | 42 |
| <b>control chart</b> review the control chart                     | 33 |
| <b>control limit</b> distinguish control limits                   | 33 |

# Find a term and its context

The page number leads to the definition and a typical word combination. Repeated terms have separate entries when they serve different lesson contexts.

|                               |                                 |    |
|-------------------------------|---------------------------------|----|
| <b>control signal</b>         | respond to a control signal     | 33 |
| <b>corner condition</b>       | evaluate corner conditions      | 24 |
| <b>corrective maintenance</b> | complete corrective maintenance | 50 |
| <b>critical dimension</b>     | measure critical dimension      | 14 |
| <b>cross-contamination</b>    | prevent cross-contamination     | 42 |
| <b>CVD</b>                    | evaluate CVD film properties    | 23 |
| <b>cycle time</b>             | state fabrication cycle time    | 69 |
| <b>data acceptance</b>        | obtain foundry data acceptance  | 69 |
| <b>defect classification</b>  | refine defect classification    | 41 |
| <b>defect density</b>         | report defect density           | 41 |
| <b>defect review</b>          | complete defect review          | 41 |
| <b>delamination</b>           | inspect for delamination        | 60 |
| <b>delivery commitment</b>    | qualify the delivery commitment | 69 |
| <b>denominator</b>            | state the denominator           | 32 |
| <b>deposition</b>             | qualify a deposition step       | 23 |
| <b>depth of focus</b>         | assess depth of focus           | 15 |
| <b>design rule</b>            | check design rules              | 68 |
| <b>dicing</b>                 | complete wafer dicing           | 6  |
| <b>die</b>                    | identify the die                | 5  |
| <b>dielectric</b>             | deposit a dielectric            | 6  |
| <b>dishing</b>                | measure dishing                 | 24 |
| <b>disposition</b>            | obtain lot disposition          | 6  |
| <b>DRC</b>                    | resolve DRC findings            | 68 |
| <b>edge cluster</b>           | investigate an edge cluster     | 41 |

# Find a term and its context

The page number leads to the definition and a typical word combination. Repeated terms have separate entries when they serve different lesson contexts.

|                                 |                                     |    |
|---------------------------------|-------------------------------------|----|
| <b>electrical yield</b>         | report electrical yield             | 32 |
| <b>endpoint detection</b>       | verify endpoint detection           | 24 |
| <b>engineering change order</b> | approve an engineering change order | 69 |
| <b>engineering lot</b>          | plan an engineering lot             | 69 |
| <b>equipment state</b>          | report equipment state              | 50 |
| <b>erosion</b>                  | assess erosion                      | 24 |
| <b>etch bias</b>                | evaluate etch bias                  | 14 |
| <b>etch selectivity</b>         | assess etch selectivity             | 23 |
| <b>exposure dose</b>            | record exposure dose                | 14 |
| <b>exposure history</b>         | reconstruct exposure history        | 42 |
| <b>exposure latitude</b>        | estimate exposure latitude          | 15 |
| <b>fab</b>                      | move material through the fab       | 5  |
| <b>fabless company</b>          | support a fabless customer          | 68 |
| <b>failure analysis</b>         | initiate failure analysis           | 60 |
| <b>FEOL</b>                     | complete FEOL processing            | 5  |
| <b>film stress</b>              | measure film stress                 | 23 |
| <b>final test</b>               | complete final test                 | 60 |
| <b>flip-chip attachment</b>     | evaluate flip-chip attachment       | 59 |
| <b>focus offset</b>             | evaluate focus offset               | 14 |
| <b>focus-exposure matrix</b>    | analyze a focus-exposure matrix     | 15 |
| <b>foundry</b>                  | coordinate with the foundry         | 68 |
| <b>GDSII</b>                    | deliver a GDSII dataset             | 68 |
| <b>HAST</b>                     | schedule HAST                       | 60 |
| <b>HTOL</b>                     | report HTOL status                  | 60 |

# Find a term and its context

The page number leads to the definition and a typical word combination. Repeated terms have separate entries when they serve different lesson contexts.

|                                                                |    |
|----------------------------------------------------------------|----|
| <b>IDM</b> work with an IDM                                    | 68 |
| <b>inspected area</b> state the inspected area                 | 41 |
| <b>inspection detection</b> review inspection detections       | 41 |
| <b>inspection sensitivity</b> confirm inspection sensitivity   | 42 |
| <b>integration trade-off</b> explain the integration trade-off | 24 |
| <b>interconnect</b> form an interconnect                       | 5  |
| <b>killer defect</b> assess killer-defect relevance            | 41 |
| <b>line-edge roughness</b> assess line-edge roughness          | 15 |
| <b>line-width roughness</b> assess line-width roughness        | 15 |
| <b>lithography</b> review lithography performance              | 14 |
| <b>lot</b> identify the lot                                    | 6  |
| <b>lot hold</b> place a lot on hold                            | 6  |
| <b>LVS</b> close an LVS mismatch                               | 68 |
| <b>mask data preparation</b> schedule mask data preparation    | 69 |
| <b>mask set</b> release the mask set                           | 69 |
| <b>matching criterion</b> apply the matching criterion         | 51 |
| <b>measurement bias</b> assess measurement bias                | 15 |
| <b>measurement recipe</b> confirm the measurement recipe       | 14 |
| <b>measurement system</b> assess the measurement system        | 33 |
| <b>metal layer</b> identify the metal layer                    | 5  |
| <b>metrology</b> review metrology consistency                  | 32 |
| <b>moisture sensitivity</b> assess moisture sensitivity        | 59 |
| <b>MOL</b> review MOL interfaces                               | 5  |
| <b>mold compound</b> qualify the mold compound                 | 59 |

# Find a term and its context

The page number leads to the definition and a typical word combination. Repeated terms have separate entries when they serve different lesson contexts.

|                                                                         |    |
|-------------------------------------------------------------------------|----|
| <b>monitor wafer</b> review monitor-wafer results                       | 51 |
| <b>MTBF</b> report MTBF                                                 | 51 |
| <b>MTTR</b> define the MTTR basis                                       | 51 |
| <b>nominal setting</b> test the nominal setting                         | 24 |
| <b>nuisance detection</b> filter nuisance detections                    | 41 |
| <b>OASIS</b> verify the OASIS file                                      | 68 |
| <b>offset correction</b> assess offset correction                       | 51 |
| <b>optical proximity correction</b> review optical proximity correction | 15 |
| <b>overetch</b> assess overetch effects                                 | 24 |
| <b>overlay</b> measure overlay                                          | 15 |
| <b>package assembly</b> begin package assembly                          | 6  |
| <b>package construction</b> identify package construction               | 59 |
| <b>package substrate</b> specify the package substrate                  | 59 |
| <b>particle excursion</b> investigate a particle excursion              | 42 |
| <b>particle-like signature</b> classify a particle-like signature       | 41 |
| <b>pass count</b> preserve the pass count                               | 32 |
| <b>pattern fidelity</b> assess pattern fidelity                         | 15 |
| <b>PDK</b> confirm the PDK revision                                     | 68 |
| <b>percentage point</b> report percentage-point change                  | 32 |
| <b>photoresist</b> coat the photoresist                                 | 14 |
| <b>pilot lot</b> evaluate a pilot lot                                   | 32 |
| <b>pitch</b> specify feature pitch                                      | 15 |
| <b>post-develop</b> review post-develop CD                              | 14 |
| <b>post-etch</b> compare post-etch measurements                         | 14 |

# Find a term and its context

The page number leads to the definition and a typical word combination. Repeated terms have separate entries when they serve different lesson contexts.

|                                    |                                    |    |
|------------------------------------|------------------------------------|----|
| <b>post-process scan</b>           | compare post-process scans         | 42 |
| <b>pre-process scan</b>            | preserve the pre-process scan      | 42 |
| <b>preconditioning</b>             | document preconditioning           | 60 |
| <b>precursor</b>                   | qualify the precursor              | 23 |
| <b>preventive maintenance</b>      | schedule preventive maintenance    | 50 |
| <b>process flow</b>                | review the process flow            | 5  |
| <b>process integration</b>         | assess process integration         | 5  |
| <b>process interaction</b>         | investigate process interactions   | 24 |
| <b>process module</b>              | qualify a process module           | 6  |
| <b>process stability</b>           | assess process stability           | 33 |
| <b>process transfer</b>            | approve process transfer           | 51 |
| <b>process window</b>              | establish the process window       | 24 |
| <b>product mix</b>                 | adjust for product mix             | 32 |
| <b>PVD</b>                         | review PVD performance             | 23 |
| <b>QFN</b>                         | review the QFN package             | 59 |
| <b>qualification approval</b>      | obtain qualification approval      | 60 |
| <b>qualification by similarity</b> | assess qualification by similarity | 60 |
| <b>qualification plan</b>          | follow the qualification plan      | 60 |
| <b>qualification vehicle</b>       | identify the qualification vehicle | 60 |
| <b>qualification wafer</b>         | evaluate a qualification wafer     | 51 |
| <b>recipe release</b>              | obtain recipe release              | 24 |
| <b>recipe revision</b>             | confirm the recipe revision        | 50 |
| <b>reference lot</b>               | identify the reference lot         | 32 |
| <b>relative change</b>             | calculate relative change          | 32 |

# Find a term and its context

The page number leads to the definition and a typical word combination. Repeated terms have separate entries when they serve different lesson contexts.

|                                                                    |    |
|--------------------------------------------------------------------|----|
| <b>release status</b> confirm release status                       | 51 |
| <b>remeasurement</b> request comparable remeasurement              | 15 |
| <b>respin</b> evaluate the need for a respin                       | 69 |
| <b>reticle</b> identify the reticle                                | 14 |
| <b>risk production</b> assess risk-production approval             | 69 |
| <b>route</b> follow the approved route                             | 6  |
| <b>rule deck</b> identify the rule deck                            | 68 |
| <b>sampling site</b> align sampling sites                          | 14 |
| <b>scheduled downtime</b> plan scheduled downtime                  | 50 |
| <b>seasoning</b> verify seasoning status                           | 51 |
| <b>setpoint</b> record the setpoint                                | 50 |
| <b>sign-off</b> complete design sign-off                           | 68 |
| <b>SiP</b> evaluate a SiP design                                   | 59 |
| <b>size threshold</b> state the size threshold                     | 42 |
| <b>slurry</b> qualify the CMP slurry                               | 24 |
| <b>solder joint</b> inspect solder joints                          | 59 |
| <b>spatial signature</b> compare spatial signatures                | 41 |
| <b>SPC</b> apply SPC                                               | 33 |
| <b>special-cause variation</b> investigate special-cause variation | 33 |
| <b>specification limit</b> check specification limits              | 33 |
| <b>step coverage</b> evaluate step coverage                        | 23 |
| <b>stratification</b> stratify by product                          | 32 |
| <b>subgroup</b> define rational subgroups                          | 33 |
| <b>surface contamination</b> investigate surface contamination     | 42 |

# Find a term and its context

The page number leads to the definition and a typical word combination. Repeated terms have separate entries when they serve different lesson contexts.

|                                                               |    |
|---------------------------------------------------------------|----|
| <b>suspect interval</b> bound the suspect interval            | 42 |
| <b>tape-out</b> plan the tape-out                             | 68 |
| <b>test coverage</b> assess qualification test coverage       | 60 |
| <b>tested population</b> define the tested population         | 32 |
| <b>thermal cycling</b> complete thermal cycling               | 60 |
| <b>thickness nonuniformity</b> report thickness nonuniformity | 23 |
| <b>thin film</b> characterize the thin film                   | 23 |
| <b>tool matching</b> assess tool matching                     | 50 |
| <b>traveler</b> update the traveler                           | 6  |
| <b>TSV</b> characterize TSV connections                       | 59 |
| <b>underfill</b> assess underfill integrity                   | 59 |
| <b>unscheduled downtime</b> investigate unscheduled downtime  | 50 |
| <b>utilization</b> state the utilization denominator          | 50 |
| <b>via</b> inspect a via                                      | 5  |
| <b>wafer</b> process a wafer                                  | 5  |
| <b>wafer fabrication</b> track wafer fabrication              | 5  |
| <b>wafer ID</b> confirm the wafer ID                          | 6  |
| <b>wafer map</b> examine the wafer map                        | 41 |
| <b>wafer sort</b> schedule wafer sort                         | 6  |
| <b>wafer start</b> confirm the wafer start                    | 69 |
| <b>waiver</b> request a defined waiver                        | 69 |
| <b>WIP</b> report WIP status                                  | 6  |
| <b>wire bonding</b> qualify wire bonding                      | 59 |
| <b>yield learning</b> support yield learning                  | 33 |

# Original cases. Clear boundaries.

These are original fictional teaching conversations, not transcripts of real workers. Names, figures, organizations, and incidents are invented. Sources support terminology and communication aims, not the fictional results or an endorsement of this book.

## **[1] ASML. How Microchips Are Made.**

[Open the original source](#)

Background on wafer processing, layers, lithography, and industry roles. The cases do not specify an actual fabrication flow or reproduce an equipment recipe. Checked 30 September 2026.

## **[2] NIST/SEMATECH. Engineering Statistics Handbook, Section 6.3.1: What Are Control Charts?**

[Open the original source](#)

Background on monitoring process behavior. The yield comparisons use invented data and do not claim a completed statistical process study. Checked 30 September 2026.

## **[3] Texas Instruments. Reliability.**

[Open the original source](#)

An industry example of reliability and qualification practice. The fictional three-test plan is not asserted as a universal qualification requirement. Checked 30 September 2026.

## **[4] SkyWater SKY130 PDK Documentation. Process Design Rules.**

[Open the original source](#)

An open example of process-specific design rules, layer definitions, and physical verification terminology. The tape-out case does not use or approve a real design. Checked 30 September 2026.

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## **Phil Stilwell**

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